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SynapSure: Mapping Brain Health

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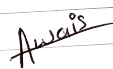
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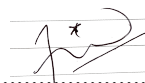
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Abstract

In our project “SynapSure: Mapping Brain Health”, we aim to revolutionize brain health diagnostics. Using various brain scan datasets we develop state-of-the-art diagnostic platform able to accurately detect both diseased states such as tumors, hemorrhages, stroke and also healthy brains. Specifically, our project relies on using advanced machine learning approaches in order to achieve high-accuracy brain scan classification. We have a user-convenient interface as well as thorough model evaluation. We use datasets to obtain information and ensure precision. Validation of data, interface improvement, testing, and implementation in real life environment, which leads to efficient management of the diagnosis and planning of treatment. These benefits may include empowering healthcare providers, facilitation of decision making on treatment and even reduction in cost of healthcare. This includes a comprehensive screening tool, and an improved early detection for brain related conditions, leading to better quality of patient care. “SynapSure: Mapping Brain Health” is an important move in healthcare technology because it brings together machine learning with healthcare.

Executive Summary

The brain is one of the most crucial parts of the human body. “SynapSure: Mapping Brain Health” project aims at developing a new solution for measuring brain health status. Brain scan based detection and classification of brain conditions involves the use of multiple brain scan datasets with employing machine learning models and methods to classify brain scans on brain tumor, brain hemorrhage, brain stroke and a healthy brain. These objectives include proper condition detection, efficient feature extraction from scan images, user-centered interface design and an extensive model evaluation. Specialized datasets for every brain condition and healthy brain scans are data sources.

The project includes validation, making well-defined user interfaces, testing and implementing it in real life to make early detection and better treatment. The planned benefits are meant to provide a good way of sorting for health professionals. This will help in finding better treatment choices and less costly treatments. The project’s outcomes include: a comprehensive brain diagnostic tool, a user-friendly interface for better early detection and treatment planning of various medical conditions related to the brain which eventually enhance patient care quality.

The thing which is important in this project is the use of modern brain scan tools. These include different image recognition methods. The unique way to handle the project is by integrating brain scans together with really smart machine learning systems. These models are meticulously designed to classify brain scans into distinct categories: brain tumors, brain hemorrhages, brain strokes, and a healthy brain scans. The anticipated benefits of the “SynapSure: Mapping Brain Health” project is important. This project wants to give doctors a strong diagnostic tool. This will help improve treatment choices, lessen medical expenses and mostly better the end results for patients. We are determined that the results of this project will be very wide in their effects.

The SynapSure project will utilize the EfficientNet model which achieves a very good accuracy of 98.17% as compare to other models. This model is chosen for its equilibrium between accuracy and computational efficiency. EfficientNet’s adaptability to diverse datasets and fine-tuning capabilities make it an optimal choice for precise classification of brain conditions. After integration of robust outlier identification and “Unknown” class handling, SynapSure becomes a reliable and effective tool for healthcare professionals in diagnosing and understanding brain conditions.

The project also knows about possible problems, like how difficult it can be to understand brain scan. It is a huge job to get lots of training information for machine learning models to work right. And, very important is the need to protect patient details and keep their data away from bad people. Firstly, the SynapSure project is being worked on which will use classifications models and if needed fix them.

Moreover, “SynapSure: Mapping Brain Health” project involves refining existing classification models and conducting thorough literature reviews on previous work in the field which ensures a comprehensive approach to addressing brain health diagnostics. This multifaceted approach aims to improve innovation in healthcare technology and enhances healthcare decision making through the integration of machine learning into medical practice.

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Chapter 1 Introduction

The word is unique, it mixes the words “**SynapSure**” which means synapse, which is confidence, and sureness which is safety in brain health analysis. This project, “**SynapSure**” It is called, “Mapping Brain Health,” and is meant to help in the building of a modern medical diagnostic aid. It utilizes multiple brain scan data sets to differentiate between several brain conditions such as brain tumor, traumatic brain injury, brain hemorrhage, brain stroke and healthy brain.

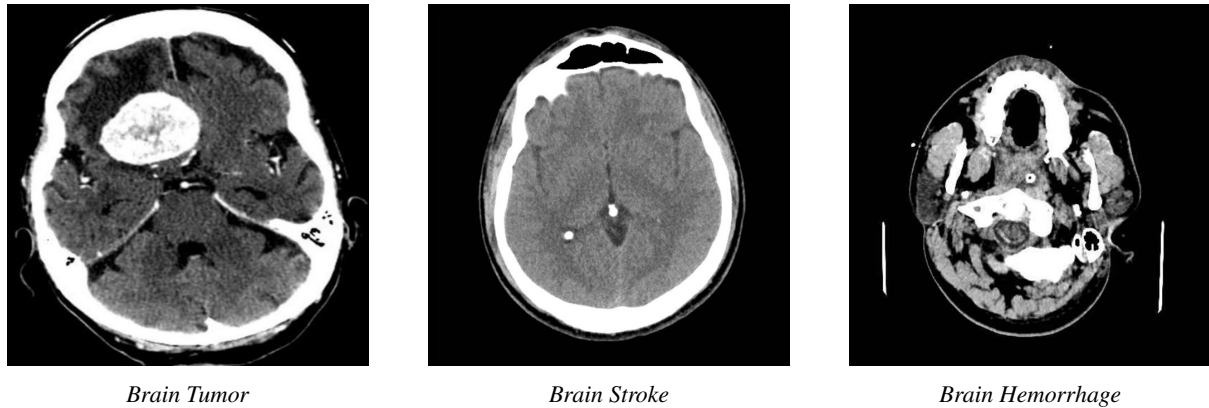


Figure 1.1: Abnormal Brain condition Scans

These are the three classes in which unhealthy brain scans are classified. Brain tumor is the abnormal growth of cells within the brain. Brain stroke is the reduced blood flow or ischemia in the brain and Brain Hemorrhage is the abnormal accumulations of blood in the brain

This effort rests upon the significance of brain health because an individual’s health, and by implication life quality, can be greatly influenced by this. By harnessing the power of modern machine learning approaches, this project is intended to provide fine-grained interpretation and classification of brain scans into appropriate diagnostic categories for early intervention and therapy planning.

The system of classifying brain health status will use brain scan images as an input provided by the users. The system will assure that the brain is healthy if it is found health. The system will also diagnose the exact condition and suggest appropriate medication, as well, in cases of abnormalities that were noted. Healthcare managers are enabled by such a function to make proper informed healthcare decisions. In summary, “SynapSure: ‘Making sense from Sensing: Mapping Brain Health” tackles this important issue by offering a novel approach that incorporates state-of-the-art technology and a thorough understanding of brain disorders.

1.1 Purpose of this Document

The purpose of this document in the ‘SynapSure: The ‘Mapping Brain Health’ program is intended to tackle the issue of brain health. The need for precise and early detection of life-threatening cerebral conditions, including brain tumors, strokes, and brain hemorrhage. Brain tumors, strokes, brain hemorrhages cause dangerous disturbances of important brain functions and result in cognitive deficits, neurological deficits, tissue damage, temporary or permanent paralysis, and increased intracranial pressure which could cause severe brain damage and Quick diagnosis and treatment helps boost survival rates while reducing the long term effects associated with TB. In order, to do this our project aims at producing a sophisticated medical diagnostic device which applies artificial intelligence methods, to provide doctors with a strong medical resource for patient care. Our process, system configuration, trials, results, and evaluation on how to manage the risk of the mentioned problems at an early stage are the scope of this report.

1.2 Intended Audience

The intended audience for the “SynapSure: Mapping Brain Health” project and its associated reports will primarily include:

- Healthcare practitioners, including neurologists, radiologists, and medical researchers, who will use the diagnostic tool for brain health assessments, early detection, and treatment planning.
- Hospitals, clinics, and medical facilities interested in adopting advanced diagnostic technology to improve patient care and outcomes.
- Scientists and researchers in the field of neurology and medical imaging who may find value in the project’s methodologies and outcomes for further research.
- Students and faculty members in healthcare-related fields, particularly those focusing on medical technology, diagnostics, and machine learning applications in medicine.
- Companies or professionals in the healthcare technology sector looking for innovative solutions to enhance their existing products or services.
- Organizations responsible for healthcare policies and regulations, as the project’s outcomes may have implications for healthcare standards and practices.
- Individuals seeking information on advancements in brain health diagnostics and early detection methods for their own knowledge or potential healthcare needs.

By considering these diverse groups, the project aims to reach a broad audience interested in improving brain health diagnostics and contributing to better patient care.

1.3 Definitions, Acronyms, and Abbreviations

Important definitions, acronyms, and abbreviations used in this chapter are as follow:

Intracranial pressure: The pressure exerted by cerebrospinal fluid within the skull, vital for brain function and shape.

Cognitive impairment: Condition where an individual experiences difficulties with memory, thinking, learning, and problem-solving.

Brain Tumor: Abnormal growth of cells within the brain **Stroke:** Medical condition that occurs when the blood supply to a part of the brain is interrupted or reduced.

Brain Hemorrhage: Bleeding that occurs within the brain tissue or the surrounding membranes.

1.4 Conclusion

The SynapSure project aims to develop an advanced diagnostic platform for brain health, utilizing machine learning and diverse brain scan datasets. This research holds the goal of improving brain health diagnostics to enhance patient outcomes. Chapter 1 provides an overview of the project's objectives and context. The primary objective of SynapSure is to offer accurate condition detection, extract relevant features from brain scans, and create a user-friendly interface for healthcare professionals. As discussed in the upcoming chapters, the project explores related work and investigates various methodologies, aligning with the broader mission of advancing brain health diagnostics.

Chapter 2 Project Vision

In this section, we outline the project's overarching goals, objectives, scope and the sustainable development milestones it aims to attain.

2.1 Problem Domain Overview

“SynapSure: Mapping Brain Health” focuses on advanced brain health diagnostics, utilizing machine learning to analyze brain scan data. The system collects brain imaging data and will train models to distinguish between critical conditions like brain tumors, strokes and brain hemorrhages and healthy brain condition. It incorporates real-time assessment and rigorous data training which aims to empower healthcare professionals with accurate early detection for improved patient care.

2.2 Problem Statement

The imperative need for accurate and early diagnosis of the serious brain diseases, including brain tumors, strokes and brain hemorrhages, gives birth to the “SynapSure: Mapping Brain Health” project. These conditions need to be detected early and correctly classified before one may plan effective treatment strategies which could in turn lead to saving lives. This gap in brain health diagnostics call for new medical diagnostic equipment that use machine learning to improve the precision of brain scans a matter of extreme importance within healthcare.

2.3 Problem Elaboration

The quality of life and well-being ultimately depends on brain health. It is important to note that this is posing challenges in brain health diagnosis at this moment. Such life-threatening conditions like strokes, brain hemorrhages and brain tumors are not diagnosed at early stages. There is a big gap that the current diagnostic methods have not been able to fill in the healthcare. These methods are less accurate and slow leading to delayed intervention in patient care which has led to many deaths warranting further research.

An example is imaging a patient with an unrecognized brain tumor. The tumour becomes undetected and interrupts normal functioning of vital brain parts incurring irreparable impairment, cognitive degeneration and sometimes death. In such cases, early detection and accurate classification may make the difference between life and death. This unfortunate situation can be greatly improved by the “SynapSure: Mapping Brain Health” project. The research paper seeks to design an automatic system in the future that will be based on the latest machine learning techniques which will be able to diagnose and

classify these fatal cerebral abnormalities. By implementing this sophisticated mechanism we aim to reduce the detrimental effects of brain illnesses on patients in a bid to avoid loss of lives.

Essentially, this issue goes beyond merely technical but humanitarian concern. This project is urgent and has the potential to transform the landscape of brain health diagnose which gives hope and brighter prospect to many people facing this life changing conditions.

2.4 Goals and Objectives

The goals of “SynapSure” are as follows:

- To promote the use of Artificial intelligence in the medical field.
- To make it less costly for the people to get the idea of their condition via the scans.
- To help doctors validate their diagnosis.

The primary objectives of the project are as follows:

- To develop a machine learning-based platform capable of accurately detecting brain tumors, traumatic brain injuries, brain hemorrhages, brain strokes and a healthy brain using brain MRI scans.
- To differentiate between various brain conditions by training datasets on different models.
- Perform validation from the output.
- To create an intuitive and user-friendly interface for medical professionals to interact with the diagnostic tool and interpret results effectively.
- To evaluate the performance of the developed model using appropriate metrics and comparing it against existing diagnostic methods.
- To suggest easy and fast diagnosis and the treatment suggestions along with the possible risks.

By achieving these goals, this project aims to develop an accurate and effective brain diagnosis platform for effortless detection of brain diseases along with recommendations.

2.5 Project Scope

The project scope for “SynapSure: Mapping Brain Health” encompasses the following major phases:

2.5.1 Project Vision

Our primary objective is to revolutionize the field of brain health diagnosis through the use of advanced technology. We prioritize improving precision and identifying potential problems at an early stage. Our goal is to transform the process of diagnosing brain-related disorders by implementing innovative techniques and technologies, thereby providing quick and accurate results. This dedication to enhancing patient results is driven by a commitment to boosting brain health through a proactive strategy. We aim to revolutionize the approach to medical diagnosis of brain.

2.5.2 Exploratory / Research Phase

- Exploring existing algorithms and techniques for brain scan analysis evaluating their effectiveness and suitability for the goals of project.
- Identify key features and parameters used by medical professionals in brain health diagnostics, understanding the nuances of accurate condition classification.

2.5.3 Implementation Phase

- Create an engaging and easy-to use useer interface.
- Gather diverse brain scan datasets and develop machine learning models capable of accurately classifying brain conditions including brain tumors, strokes, and brain hemorrhages as well as healthy brain states.
- Implement user interaction capabilities, allowing individuals to input brain scan data for analysis.
- Apply advanced Image data augmentation technique etc, to enhance the quality of input data.
- Providing clear insights into brain health conditions, and produce corresponding Recommendations for treatment of classified brain condition.

The project's scope extends to the comprehensive evaluation and validation of the developed system, ensuring its reliability and effectiveness in real-world healthcare scenarios.

2.6 Sustainable Development Goal (SDG)

The primary SDG addressed by the “SynapSure: Mapping Brain Health” project is:

2.6.1 SDG 3: Good Health and Well-being

Our project aligns with SDG 3 by focusing on the improvement of brain health diagnostics, early detection, and precise classification of life-threatening brain conditions. By enhancing healthcare professionals' abilities to provide timely interventions and treatment planning, we contribute to the goal of ensuring good health and well-being for all.



Figure 2.1: Sustainable Development Goal 3

This figure ensures healthy lives and promotes well-being for all which focuses on access to essential healthcare, disease prevention and mental health.

Additionally, the project indirectly aligns with:

2.6.2 SDG 16: Peace, Justice, and Strong Institutions

Through the prevention of misdiagnoses and the promotion of unbiased and accurate brain health assessments, our project supports the establishment of strong and just institutions by ensuring fair and equitable access to healthcare and diagnostic services.



Figure 2.2: Sustainable Development Goal 16

Goal 16 promotes peaceful and inclusive societies by ensuring access to justice, building effective, accountable institutions and fostering the rule of law.

2.7 Constraints

- Limited access to diverse and high-quality brain scan datasets may pose a constraint during model development and testing.

2.8 Business Opportunity

- The project capitalizes on the growing demand for advanced medical diagnostic tools, particularly in the field of brain health diagnostics.
- SynapSure has the potential to significantly enhance healthcare by improving brain condition detection, leading to better patient care and outcomes.

2.9 Stakeholders Description/ User Characteristics

- Primary users include neurologists, radiologists, and other medical experts who will use SynapSure for accurate brain health diagnostics.
- Individuals seeking early and precise brain condition detection and treatment.
- Hospitals, clinics, and healthcare facilities looking to enhance their diagnostic capabilities and provide better patient care.

2.10 Conclusion

Chapter 2 outlines the comprehensive vision of the project. The primary objective of the "SynapSure: Mapping Brain Health" is to transform the field of brain health diagnostics using cutting-edge technology. There is an urgent requirement to detect brain conditions such as tumors and strokes accurately and at an early stage. The goal is to use machine learning to identify various brain conditions. The section underscores the correlation with Sustainable Development Objective 3, concentrating on the promotion of good health and well-being. Furthermore, it highlights limitations due to limited availability of varied data sets and acknowledges a commercial opportunity arising from the increasing demand for sophisticated medical diagnostic technologies. The sentence is describing stakeholders, who are medical experts and individuals seeking early detection. The statement emphasizes the influence these stakeholders can have on healthcare facilities. The project's objectives and stages are defined, providing a basis for its impactful contribution to the field of brain health diagnosis.

Chapter 3 Literature Review / Related Work

The literature review in this research serves as a vital tool to define the project's scope. We comprehensively examined image classification methods, summarizing each article's key points. In these summaries, we highlighted each paper's objectives and methodologies. Furthermore, we assessed the strengths, weaknesses, and discussed how these models can benefit us.

3.1 Definitions, Acronyms, and Abbreviations

FAHS-SVM: Fully Automatic Heterogeneous Segmentation using Support Vector Machine

MRI: Magnetic resonance imaging

TSS: Time Since Stroke

AE: Auto Encoder

LR: Logistic Regression

DWI-FLAIR: Diffusion-Weighted Imaging–Fluid-Attenuated Inversion Recovery

CNN: Convolutional Neural Network

K-NN: K Nearest Neighbor

AI: Artificial Intelligence

FF-ANN: Feed Forward - Artificial Neural Network

RSwithRF: Random Subspace with Random Forest

RSwithBN: Random Subspace with Bayesian Network

DWT: Discrete Wavelet Transformation

FP-ANN: Feed-forward Back-propagation Artificial Neural Network

SVM: Support Vector Machine

2D: 2 Dimension

GPU: Graphics Processing Unit

CT: Computed Tomography

Lung Nodule: A lung nodule is a small, round or oval-shaped mass or spot in the lung that is often detected through medical imaging, such as a chest X-ray or CT scan

3.2 Detailed Literature Review

Our topic focuses on brain scan classification thus we have done vast research in this area. Image classification has three major parts that are image data preprocessing, feature extraction and classification.

Thus, we have gathered a vast amount of literature review regarding these topics.

3.2.1 Brain Tumor Identification and Classification of MRI images using deeplearning techniques [1]

3.2.1.1 Summary of the research item

The research paper introduces a novel approach called FAHS-SVM for the automated identification and segmentation of brain tumors. The method aims to achieve accuracy levels comparable to manual segmentation, addressing the challenge of inter-observer variability. By leveraging both image structure hierarchy and statistical classification, the technique effectively identifies spatially small and consistent tumor areas, serving as a robust guide for subsequent segmentation. Experimental results on multi-parametric Magnetic Resonance images demonstrate the method's promise in accurately and quickly locating brain tumors, with an impressive 98.51% accuracy in distinguishing abnormal and normal tissues. Overall, this research holds significant potential for aiding radiologists and clinical experts in diagnosing and supporting clinical decision systems in the field of MRI brain tumor detection.

3.2.1.2 Critical analysis of the research item

Strengths:

- High reported accuracy (98.51%).
- Focus on full automation for tumor detection.
- Combination of image structure and classification techniques.
- Emphasis on clinical relevance.

Weaknesses:

- Lack of technical details about the method.
- Limited information on the dataset and validation.
- Integration into clinical workflows not discussed.

3.2.1.3 Relationship to the proposed research work

The “SynapSure: Mapping Brain Health” project shares similarities with the research work on “Brain Tumor Identification and Classification of MRI images using deep learning techniques.” Both projects involve the application of machine learning and deep learning techniques for the detection and classification of brain-related medical conditions. While the specific focus of the research work is on brain

tumors in MRI images, “SynapSure” aims to develop a broader diagnostic platform that covers various brain conditions, including brain tumors, hemorrhages, strokes, and healthy brain assessment. Both projects leverage medical imaging data to advance early diagnosis and treatment planning, contributing to improved patient care and healthcare decision-making.

3.2.2 A Machine Learning Approach for Classifying Ischemic Stroke Onset Time From Imaging [2]

3.2.2.1 Summary of the research item

In the research titled “A Machine Learning Approach for Classifying Ischemic Stroke Onset Time From Imaging,” the study aimed to classify the time since the onset of ischemic stroke (TSS) using machine learning techniques applied to medical imaging data. The methodology involved determining optimal AE architectures for different coupling patches, generating deep feature maps from these AEs, and training classifiers using both baseline imaging features and deep AE features. Various subgroup and sensitivity analyses were conducted to assess the robustness of the approach. The study found that machine learning models particularly the optimal LR classifier could outperform the traditional DWI-FLAIR mismatch method for TSS classification. Additionally, correlation analysis demonstrated that deep autoencoder features captured complex information contributing to improved classification performance.

3.2.2.2 Critical analysis of the research item

Strengths:

- The research introduces a novel machine learning approach for classifying ischemic stroke onset time (TSS) from medical imaging data, potentially improving upon existing methods.
- The study conducted sensitivity and subgroup analyses to assess the reliability and applicability of the proposed approach under various conditions.
- The correlation analysis provided some insight into the deep learning features, making the model more interpretable for clinical use.
- Emphasis on clinical relevance.

Weaknesses:

- The study used a relatively small dataset (131 patients), which could limit the generalizability of the findings.
- Deep learning models, though effective, can be challenging to interpret, potentially hindering their clinical adoption.

- Clinical variables, such as patient age, were not considered in the classification, which might have improved accuracy and clinical relevance

Overall, the research offers a promising approach for TSS classification but requires further validation on larger datasets and additional clinical variables to enhance its practical utility.

3.2.2.3 Relationship to the proposed research work

“SynapSure: Mapping Brain Health” aligns with the above research work by leveraging machine learning techniques and datasets to classify various brain conditions, similar to the brain tumor detection and intracranial hemorrhage projects mentioned. It aims to provide a user-friendly interface for brain health assessment, contributing to early diagnosis and improved patient care, consistent with the goals of the referenced projects in medical imaging and diagnosis.

3.2.3 Brain MRI Analysis for Alzheimer’s Disease Diagnosis Using CNN-Based Feature Extraction and Machine Learning [3]

3.2.3.1 Summary of the research item

This article is covering why we need an early diagnosis of Alzheimer’s disease, considering its status as the most common form of dementia and a significant cause of death. Traditional diagnosis techniques use handcrafted feature extraction methods, which was very complicated. To streamline this process, the study introduces a pre-trained CNN model, ResNet50, as an automatic feature extraction method for diagnosing Alzheimer’s disease through MRI images. The paper shows the accuracy of the proposed model, ranging from 85.7% to 99%, way better than existing methods. This research presents a significant advancement in Alzheimer’s disease diagnosis, potentially improving patient outcomes through early and precise detection.

3.2.3.2 Critical analysis of the research item

Strengths:

- The study exploring Alzheimer’s disease diagnosis through CNN based feature extraction and machine learning integration expresses innovation in diagnosis.
- By utilizing CNN for MRI images, the study presents an almost accurate Alzheimer’s diagnosis.

Weaknesses:

- The research focus on Alzheimer’s disease diagnosis raises concerns about its applicability to a broader field of neurological disorders, such as tumors and strokes.

- Study lacks specific details regarding the architectures of the CNNs used and the machine learning algorithms integrated creates a gap in understanding the research methodology.

3.2.3.3 Relationship to the proposed research work

Our research and this article are performing the same operation of classifying Alzheimer disease but on different data sets. Both studies involve the application of machine learning techniques to medical imaging data, emphasizing the importance of accurate diagnosis. By understanding their approach, we can adapt similar techniques for analyzing MRI images related to brain cancer, enhancing the efficiency and precision of our own machine learning models for brain diseases detection.

3.2.4 Transfer Learning for Automatic Brain Tumor Classification Using MRI Images [4]

3.2.4.1 Summary of the research item

Brain tumor segmentation and classification using MRI through deep learning techniques is very challenging. Also, manual classification with human assistance can lead to improper prediction and diagnosis. So, the study proposes a deep learning model for brain tumor classification, employing CNNs based on transfer learning. Three pre-trained CNN architectures, called: ResNet, Xception, and MobileNet-V2, are discussed. Among these, MobileNet-V2 achieves the best results with an accuracy of 98.24%. Transfer learning is used to leverage pre-trained models for this task, addressing the challenge of limited available medical data.

3.2.4.2 Critical analysis of the research item

Strengths:

- This paper used a very brilliant strategy of transfer learning to address the lack of medical data.
- The model used data augmentation, normalization, and resizing techniques for brain tumor classification, ensuring the robustness of the system.

Weaknesses:

- The model only classifies brain tumors and its scope is limited.
- We cannot classify other brain diseases as the model trained through transfer learning restricts our ability to classify other brain diseases.

3.2.4.3 Relationship to the proposed research work

In healthcare, there's a lack of medical data for training advanced deep learning models. To solve this, the study uses transfer learning, using pre-trained models for new tasks, and solving data scarcity problems. We can use this technique for better classification of diseases with limited data set.

3.2.5 An Efficient Classification of MRI Brain Images [5]

3.2.5.1 Summary of the research item

The research highlights a significant advancement in medical imaging through the utilization of MRI for diagnosing brain disorders and injuries. This paper aims to develop an efficient method for classifying MRI brain images. The research proposes a simple yet efficient method for classifying MRI brain images as normal or abnormal, brain conditions like brain tumors, brain strokes, and Alzheimer's disease. The methodology comprises four key stages: pre-processing involving scalp and skull removal, feature extraction, feature reduction, and classification. The study employs individual classifier called FF-ANN alongside hybrid classifiers called RSwithRF and RSwithBN, resulting in impressively accurate classifications 95.83%, 97.14% and 95.71% respectively. These outcomes highlight the method's potential in medical image classification.

3.2.5.2 Critical analysis of the research item

Strengths:

- The study suggested a new mechanism to differentiate MRI brain images into normal and abnormal using individual and hybrid classifiers.
- The model is based on a four-step process, data pre-processing, feature extraction, feature reduction, and classification for precise MRI brain image categorization.
- Accuracy is 97.14% which is far better than existing techniques.

Weaknesses:

- The evaluation involved only one single individual and two hybrid classifiers.
- Testing was conducted on just 70 images from a single dataset.
- Analysis focused on only three statistical features.
- The study does not compare its findings with the latest research in the field, particularly those based on deep learning techniques.

3.2.5.3 Relationship to the proposed research work

This study deals with the classification of MRI brain images, which aligns closely with our focus on utilizing MRI images for brain diseases detection. Understanding their approach to MRI image classification, even at a high level, might provide inspiration or points of comparison for our machine learning models. Considering the similarities in the use of MRI images for classification purposes, analyzing the methods presented in the paper could enhance the foundation of our research. It is important to critically observe their techniques, compare their results, and identify gaps in the context of brain cancer detection which potentially lead to more robust and accurate machine learning models for brain disease detection.

3.2.6 Classification of MRI Brain Images using k-Nearest Neighbor and Artificial Neural Network [6]

3.2.6.1 Summary of the research item

The research introduces an automated diagnosis approach on classification of MRI commonly utilized in outlining soft tissue. The method involves two main stages: feature extraction and classification. During the first stage, DWT is applied to MRI images which enables the extraction of frequency space information from non-stationary signals. The obtained features are then reduced to essential components using principal component analysis. In the classification stage, two classifiers are used: a FP-ANN and a k-NN classifier. The results demonstrate a classification success rate of 90% and 99% for FP-ANN and k-NN respectively which indicates the effectiveness of the proposed technique in automated diagnosis.

3.2.6.2 Critical analysis of the research item

Strengths:

- The research explores a various set of methodologies, enhancing the depth of analysis and comparison by employing both k-NN and ANN,
- The combination of traditional machine learning (k-NN) and deep learning (ANN) techniques allows for a broad evaluation.
- The study helps medical professionals to classify brain conditions with accuracy of 99%.

Weaknesses:

- The paper lacks an in-depth exploration of the limitations of the chosen classification methods.
- Limited information provided regarding the dataset used, such as its size, diversity, and sources.

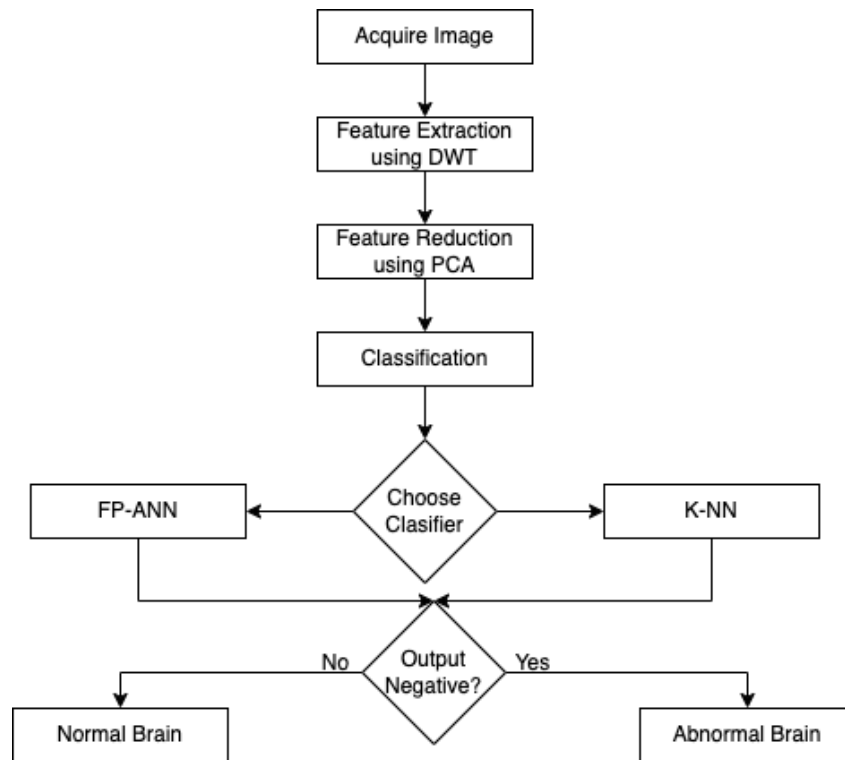


Figure 3.1: Methodology Used in Research

This figure shows the complete methodology of the research. The first step is that the image is acquired and then feature extraction is done using DWT and then feature extraction is done using PCA. We choose the classifier either FP-ANN or K-NN and then it shows the if the output is negative or not and it classifies into normal or abnormal brain

3.2.6.3 Relationship to the proposed research work

One of the end goals of our project is to classify whether the brain scan is normal or not. This project uses a K-NN classification model and has the accuracy of 99%. So we can learn from the classification techniques used in the model preparation to increase the accuracy of our project.

3.2.7 A Hybrid CNN-SVM Threshold Segmentation Approach for Tumor Detection and Classification of MRI Brain Images [7]

3.2.7.1 Summary of the research item

The main thing discussed in this research paper is that it presents a wide view on detection of brain tumor and its classification using MRI images. The two technologies or methodologies used in this are CNN and SVM. The main reason for this research was to timely treat the brain tumors and so they could be detected timely. The accuracy which was obtained from this by using the public data set (BRATs 2015) was 98.4959%.

3.2.7.2 Critical analysis of the research item

Strengths:

- The strength of this model was the combined use of CNN and SVM.
- This hybrid approach resulted in higher accuracy as they achieved an accuracy of 98%.
- This research paper provides all the pre-processing steps and the statistical functions which are used by the model.

Weaknesses:

- The weakness of this paper is that it does not discuss real world clinical validation
- There is an assumption of preprocessed data.

3.2.7.3 Relationship to the proposed research work

The hybrid approach in which CNN and SVM are working together can be used in our research and it can lead to improved accuracy. By studying this paper, we can make a model which can be more accurate than this and it will be very useful.

3.2.8 Diagnosis of Alzheimer Disease Using 2D MRI Slices by Convolutional Neural Network [8]

3.2.8.1 Summary of the research item

The focus of this research item is to focus on the concepts of deep learning and specifically convolutional neural networks for the purpose of detecting brain Alzheimer using 2D MRI. This paper gives us the importance of machine learning and early diagnosis of these diseases. The dataset was obtained from the OASIS and CNN was used to train the structure. The research aims to investigate various parameters, including the number of layers, the number of filters, and dropout rates, to optimize the performance. The results were very good, and they achieved an accuracy of 99.30% in distinguishing between Alzheimer patients and healthy individuals.

3.2.8.2 Critical analysis of the research item

Strengths:

- This research involves a novel approach which uses a 2D CNN and it had computational advantages.
- The accuracy was 99.30% which is a very high strength.

- The research provides a detailed account of the CNN architecture, parameters, and preprocessing steps used in the study, which aids in reproducibility.

Weaknesses:

- The research lacks the in-detail discussion about the size of the dataset and potential biases.
- It didn't specify the details of the dataset.

3.2.8.3 Relationship to the proposed research work

The work in this research is related to our work as it gives us research on the detection of Alzheimer using deep learning and CNN. Our goal is to classify the disease of brain or if it is a healthy brain and this research is valuable for us. The accuracy of 99.3% shows that the results are very precise, and they are of great value to us.

3.2.9 Brain Tumor Detection and Classification Using Convolutional Neural Network and Deep Neural Network [9]**3.2.9.1 Summary of the research item**

This research paper mainly focuses on the concept of machine learning and CNN for the detection of brain tumor and the automated detection of brain tumor using MRI images. This research combines the neural networks and the CNN in order to achieve a high accuracy. The accuracy which was achieved by this research and its model was 96.03%.

3.2.9.2 Critical analysis of the research item**Strengths:**

- The model was successful with a very high accuracy of 96% and it gives a deep concept of the CNN which was used to detect the brain tumor.
- It successfully differentiates between a healthy individual and a person with brain tumor.
- It detects the brain tumor early which can save lives of people.

Weaknesses:

- The research does not mention the main elements of the data set which are its size and the other main elements of the data set.
- There was no proper differentiation between the training data and the validation data.

3.2.9.3 Relationship to the proposed research work

This research is related to our research work as it gives a very higher accuracy in detecting the brain tumor and our work is also to classify between various brain diseases and it will be of great help in our own research.

3.2.10 Deep Learning Based Brain Tumor Detection and Classification [10]

3.2.10.1 Summary of the research item

The main thing discussed in this research is the detection of MRI. It mainly is the research of the development of two deep learning bases approaches. There is a fast AI deep learning library. The study is conducted using a subset of the BRATS 2018 data set, containing 1,992 Brain MRI scans. The first model achieved the accuracy of 85.95% and the second model achieved the accuracy of 95.78%.

3.2.10.2 Critical analysis of the research item

Strengths:

- The concepts of deep learning are used which have shown very precise and accurate results.
- This research gives a comparison of two deep learning models and one is more accurate but we can say that both models achieved a very high accuracy.
- The Data set which was used is well known and publicly available.

Weaknesses:

- The research mentions that use of google Colab and tesla p100 GPU which is not available in all healthcare platforms.
- The Data set is a subset of BRATS 2018 Data set and because of that the accuracy is less.
- The research also does not show any Pre-processing steps.

3.2.10.3 Relationship to the proposed research work

This research paper is related to our research as our research is also about differentiation between various brain diseases and this model shows a high accuracy in differentiating between a healthy brain and the one with brain tumor so this research is related to our research.

3.3 Application Review

There some applications we took motivation from, which are related to our work, with main functionality of image classification via ML to diagnose various diseases.

3.3.1 ABcanDroid [11]

ABcanDroid is an application for early detection of breast cancer for better diagnosis. In the application CNN implemented along with Transfer Learning for increasing the efficiency and accuracy of results. The overall accuracy is 99.58%. ABcanDroid identifies Breast Cancer using images provided by users.

First of all it converts these images into a specific color format Red-Green-Blue. Only images similar to Breast Histopathological Images are used. The system checks if an image is structurally similar to a Histopathological image using a measure called Structural Similarity Index. If it is similar, the image is used for detection. The steps involved are, splitting data for testing and training, using advanced models like ResNet for accurate classification. The system is deployed on a cloud service called Heroku, and the results are accessed through an Android app made with Flutter framework. This approach prevents the loss of accuracy.

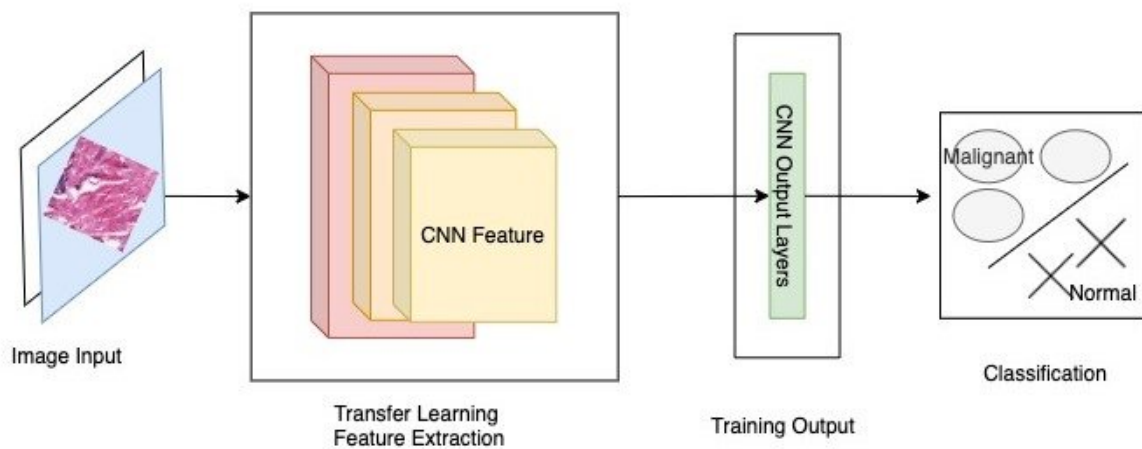


Figure 3.2: ABcanDroid Model Architecture

This figure shows the complete architecture of the ABcanDroid Model. The model takes the image as input, then it extracts the features using transfer learning and then it trains the output and classifies between normal and malignant scan

The benefits of ABcanDroid includes it can be affordable and accessible to healthcare professionals. Also it is trained on small datasets by using transfer learning techniques and the accuracy of this model is very high. The framework requires very minimal human intervention and as it is cloud integrated, so less performance load on the edge devices.

3.3.2 Optellum [12]

Optellum is a healthcare technology website that helps in lung cancer diagnosis and treatment. Optellum uses advanced artificial intelligence and machine learning techniques to the detection and diagnosis of lung cancer. Their approach involves employing AI algorithms to analyze lung scan, specifically CT scans of the lungs. CT scans provide detailed cross-sectional images of lung tissues.

Optellum's module, **the Virtual Nodule Clinic**, is designed to assist healthcare professionals in accurately diagnosing lung nodules found in these CT scans. The system takes CT scan images of the patient's lungs as input. These images are showing structures and abnormalities within the lung tissues. Optellum's AI algorithms analyze the CT scan images. By machine learning, the system identifies patterns and anomalies associated with lung cancer. This analysis includes the size, shape, density, and growth rate of lung nodules. The AI system classifies the detected nodules into different categories, distinguishing between benign (non-cancerous) and malignant (cancerous) growths. This classification is crucial for determining the appropriate course of action and treatment. Also it assesses the risk associated with identified lung nodules. It evaluates factors such as the patient's medical history, nodule characteristics, and other relevant data to determine the likelihood of malignancy.

The results and risk assessments provided by the model serve healthcare professionals for better treatment. Physicians can use this information as a basis for making decisions regarding further diagnostic tests, biopsies, or treatment options.

3.3.3 Brain Classifier Web-App [13]

"Brain Tumor Classifier Web-App" involves methods for detecting brain tumor. The model is based on CNN trained on more than 800 images of different categories of brain tumor. The classifier not only classifies between Tumor and No tumor but will also return the class of the Tumor if it is present. The user input the image of the MRI scan. This image is then preprocessed to make it similar to the input on which the model was initially trained. Once the processing is done, the image is fed to the model which then gives the user a result. The model that makes the predictions is a Convolutional Neural Network which consists of 5 Convolutional layers and 5 Max Pooling layers and it was trained on a batch size of 40 and for a total of 50 epochs. But it did classify those images correctly which had Tumor even though it wasn't accurate in telling which type of tumor it was. The model gave an accuracy of 97.5% on the training set and an accuracy of 94% on the validation set.

3.4 Literature Review Summary Table

Following is the table providing findings of past articles related to image data classification.

Table 3.1: Literature Review Summary Table

This table shows a brief summary of literature review findings. There are 4 columns. The first is the number and then there is author and then the other is the accuracy and the last column is model

No.	Author	Accuracy	Model
1	Jia, Zheshu and Chen, Deyun A [1]	98.51%	Fully Automatic Heterogeneous Segmentation using Support Vector Machine (FAHS-SVM)
2	AlSaeed, Duaa omar and Samar Fouad [3]	85.7% to 99%	Convolutional Neural Network (CNN) model, ResNet50
3	King Chung et. al [2]	92%	Logistic Regression and Autoencoders
4	Arbane et. al [4]	98.24%	CNN architectures—ResNet, Xception, and MobileNet-V2
5	Assam et. al [5]	95.83%, 97.14%, and 95.71%	Feed Forward - Artificial Neural Network (FF-ANN), Random Subspace with Random Forest (RSwithRF) and Random Subspace with Bayesian Network (RSwithBN)
6	Rajini, N Hema and Bhavani [6]	90% and 99%	Feedforward Backpropagation Artificial Neural Network (FP-ANN) and K-Nearest Neighbor (k-NN)
7	Khairandish et. al [7]	98.4959%	Convolutional Neural Network and Support Vector Machine (SVM)
8	Al-Khuzai et. al [8]	99.30%	Convolutional Neural Networks
9	Chirodip Lodh et. al [9]	96.03%	Convolutional Neural Networks
10	Dipu et. al [10]	85.95% and 95.78%	YOLOv5 and FastAI

3.5 Conclusion

The literature review presents a comprehensive overview of various models and methodologies applied in the field of brain health diagnostics, particularly in the context of brain tumor and disease detection from MRI images. The reviewed studies have leveraged a diverse range of techniques, including

Convolutional Neural Networks (CNNs), Support Vector Machines (SVMs), transfer learning, and deep feature extraction. Notably, these models have consistently demonstrated high levels of accuracy, often surpassing 90%, and, in some cases, achieving accuracy rates above 99%. These findings underscore the efficacy of machine learning and deep learning approaches in accurately identifying brain conditions from medical imaging data. As our project, “SynapSure: Mapping Brain Health,” seeks to contribute to this field, the insights gained from this literature review will inform our choice of methodologies and guide the development of a robust brain health diagnostic tool with a focus on achieving high accuracy levels for early and precise condition detection.

Moreover, after reviews some similar applications [11] [12] [13] , we noticed some important points which mainly includes user’s data privacy, hence we aim to make a diagnostic tool which will ensure to focus on user data safety by adding sign-up and login features for confidentiality. Moreover, our system is highly precise, utilizing advanced classification algorithms to enhance the reliability of disease predictions. After a precise diagnosis, our app provides users with precautionary suggestions. Our app is user-friendly, making it accessible to everyone, even those without technical and even medical expertise. These comparisons highlights how our research enhances disease detection accuracy, coverage, and user data safety in healthcare diagnostics.

Chapter 4 Software Requirement Specifications

In this section a detailed description of all related software requirements for our project is presented. Our project comprises a listing of the characteristics, functional specifications, quality standards, non-functional requirements, related assumption, hardware and software requirement, scenarios, user interface design and analysis of the risks associated with our project.

4.1 List of Features

The following features will be available in the system.

- Healthcare providers should have secure sign-up and log-in functionality.
- Compatibility for uploading different brain scan datasets like MRI images.
- Pre-processing of brain scans.
- Various trained datasets based on different models for differentiating brain tumor or stroke from hemorrhage.
- Display suggestions if the scan has any brain condition.

4.2 Functional Requirements

The following functional requirements shall be available in the system.

- The user shall be able to create an account by filling out the information required.
- The user shall be able to login to their account by providing credentials and use the software.
- The user shall be able to logout of their account.
- The user shall be able to upload their brain scans in multiple formats.
- The system shall convert the images into the required format.
- The system shall implement a machine learning algorithm to classify whether the scan is cancerous or not.
- The system shall implement machine learning algorithms for the image classification of various brain conditions (stroke, Alzheimer, and hemorrhage).
- The system shall provide a diagnostic result.
- The system shall display recommendations to the user based on their brain condition.

4.3 Quality Attributes

The project aims to incorporate the following fundamental quality attributes in order to ensure the prevention of application defects. Each attribute shall be used to evaluate the application quality and performance during operation.

4.3.1 Reliability

It is important to ensure the reliability of the project as it is designed for medical purposes. We need to make sure that the application gives the right results. It does not matter where it is used or what conditions it is working in. The product should always be accurate and trustworthy. This is important because people rely on the app to provide correct information in medical situations and we want to make sure it does that every time.

4.3.2 Maintainability

SynapSure ensures long term viability with a set of pre-trained segmentation models for brain scans analysis. While the current algorithms represent the latest advancements the system is designed to seamlessly integrate future algorithms which ensures continuous state-of-the-art diagnostic capabilities. Considering this need the application shall have enough maintainability to update the algorithms or switch the underlying technologies for efficient inferences later on.

4.3.3 Usability

The project aims to serve as a utility for general public and medical professionals to assist in their diagnostic workflow. The project must have high usability and accessibility as the audience is non technical. The usability of the project shall be evaluated with the following performance indicators.

- Consistent user interface across the application to avoid any confusion.
- Easy for new or infrequent users to learn to use the system.
- Easy for users to input data and interpret the output.

4.3.4 Correctness

Considering a medical context, it is critical that the application produces correct results and avoids false positives. In order to achieve this goal the project is committed to following the functional requirements closely. The focus is on preventing false positives which means the system should not incorrectly identify something as true when it is not true.

4.4 Non-Functional Requirements

4.4.1 Availability

- The system shall be readily available to give response to the user's requests.
- The system should be available to always detect brain diseases 24/7.

4.4.2 Reusability

- The code should be divided into modules based on the functionality so that it favors code reusability.
- The system should be able to classify brain scans on a consistent basis and should not fail after a specific period of working capability.

4.4.3 Robustness

- The system shall able to process and deal with extensive amounts of data without loss of critical information and operational capability.
- The system shall preserve the progress, in case of connection dropouts or hardware failures.

4.4.4 Security Requirements

- The system shall allow the users to have a unique email/phone number of their accounts from which the only particular user can access his account.
- The system shall not be able to share the user data with anyone.

4.4.5 Performance

- The system shall be able to give responses to the user's request in real time.

4.5 Assumptions

The assumptions made for the specification for the user to use the system are listed as:

- The user has sound knowledge of the project's purpose and utility.
- The user will input their brain scan. It is assumed that the users will have their brain scans.
- The users have basic knowledge of how to use a computer and website.

4.6 Use Cases

Following are the use cases identified in our web application.

4.6.1 Sign Up

Name	Sign Up												
Actors	User												
Summary	Users create a new account on the SynapSure platform by providing required registration information.												
Pre-Conditions	The user must fill out the whole sign up form. None of the user’s information exists in the database.												
Post-Conditions	Account is successfully created. The user is sent to the login page.												
Special Requirements	Valid email address and secure password												
Basic Flow													
<table><tr><th colspan="2">Actor Action</th><th colspan="2">System Response</th></tr><tr><td>1</td><td>User selects the 'Sign Up' option.</td><td>2</td><td>System displays the registration form.</td></tr><tr><td>3</td><td>User provides the required information, including email and password.</td><td>4</td><td>System validates the information and create account.</td></tr></table>		Actor Action		System Response		1	User selects the 'Sign Up' option.	2	System displays the registration form.	3	User provides the required information, including email and password.	4	System validates the information and create account.
Actor Action		System Response											
1	User selects the 'Sign Up' option.	2	System displays the registration form.										
3	User provides the required information, including email and password.	4	System validates the information and create account.										
Alternative Flow													
3	User enters incorrect input fields.	4-A	System displays an error message: 'Invalid Information Entered'.										
3	Any necessary input fields are omitted by the user.	4-B	System displays an error message: 'Please fill in the missing field(s)'.										

4.6.2 Login

Name		Login	
Actors		User	
Summary		Users access their accounts by providing their login credentials.	
Pre-Conditions		User must have a registered account on SynapSure. User has a valid email and password.	
Post-Conditions		User is successfully logged into the account. The user is redirected to the SynapSure dashboard.	
Special Requirements		Valid email address and password	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Login' option.	2	System displays the login screen which requests email and password.
3	User inputs a valid password and email.	4	System authenticates the user's email and password and redirects the user to SynapSure's dashboard.
Alternative Flows			
3	User enters incorrect login credentials.	4-A	System displays an error message: 'Invalid Login Credentials'.

4.6.3 Upload Brain Scan

Name		Upload Brain Scan	
Actors		User	
Summary		Users upload brain scan images into the SynapSure system. The system then prepares the scans for analysis.	
Pre-Conditions		User is registered and logged into the system. The brain scan image is in a supported format.	
Post-Conditions		The brain scan is successfully uploaded and queued for analysis.	
Special Requirements		High-resolution scan images	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Upload Scan' option.	2	System displays the upload interface.
3	User uploads the brain scan image.	4	System confirms the upload and queues the scan for analysis.
Alternative Flow			
3	Medical professional attempts to upload an unsupported file format.	4-A	System displays an error message: 'Unsupported File Format'.

4.6.4 View Analysis Result

Name		View Analysis Result	
Actors		User	
Summary		Users access and review the analysis results of uploaded brain scans.	
Pre-Conditions		User is logged into the system. A brain scan has been analyzed by the system.	
Post-Conditions		User has viewed and understood the analysis results.	
Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User selects the 'View Results' option for a specific scan.	2	System displays the analysis results.
Alternative Flow			
1	Medical professional selects a scan that is still being analyzed.	2-A	System displays a message: 'Analysis still in progress. Please check back later.'

4.6.5 Provide Feedback

Name	Provide Feedback
Actors	User
Summary	Users provide feedback on their experience with the system includes suggestions for improvements, reporting issues and general comments.
Pre-Conditions	User is registered and logged into the system.
Post-Conditions	Feedback is recorded and available for system evaluation.

Special Requirements		None	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Provide Feedback' option.	2	System displays the feedback form.
3	User fills out the feedback form and submits it.	4	System confirms receipt of the feedback.
Alternative Flow			
3	User submits an incomplete form.	4-A	System prompts the user to complete all required fields.

4.6.6 Generate Report

Name		Generate Report	
Actors		User	
Summary		After analyzing a brain scan, users can generate a detailed report that includes the diagnosis and other relevant findings.	
Pre-Conditions		User is logged in. A brain scan has been analyzed and results are available.	
Post-Conditions		A detailed diagnostic report is generated and available for review or download.	
Special Requirements		Report must be clear, concise, and medically accurate.	
Basic Flow			
Actor Action		System Response	
1	User selects the 'Generate Report' option for a specific scan.	2	System compiles the analysis results into a comprehensive report format.
3	User reviews or downloads the generated report.	4	System provides options to review on-screen or download the report in a preferred format.
Alternative Flow			
3	An error occurs during report generation.	4-A	System displays an error message and suggests retrying or contacting support.

4.7 Hardware and Software Requirements

The project employs a multitude of hardware and software services for its operations. For the purpose of developing and deploying this project following hardware and software requirements need to be fulfilled.

4.7.1 Hardware Requirements

The following are the hardware requirements required for the operation of the system:

- A computer system will be required to develop this website. Core i5 or later processors will be ideal for development of this web-based project.
- The device shall have the sufficient space available to store the brain scans.
- Internet connection is required to set up the system.

4.7.2 Software Requirements

The following are the software requirements required for the operation of the system:

- Anaconda
- Python, 3.8 or above
- Web Browser i.e. Google Chrome or Firefox
- MongoDB
- Python Libraries
- GitHub
- React
- JavaScript
- Draw.io

4.8 Graphical User Interface

This section should give the GUI dumps of each screen, with reference to the users. The navigation flow of each user is also required, and each GUI should mark the functionality/use case that it covers.

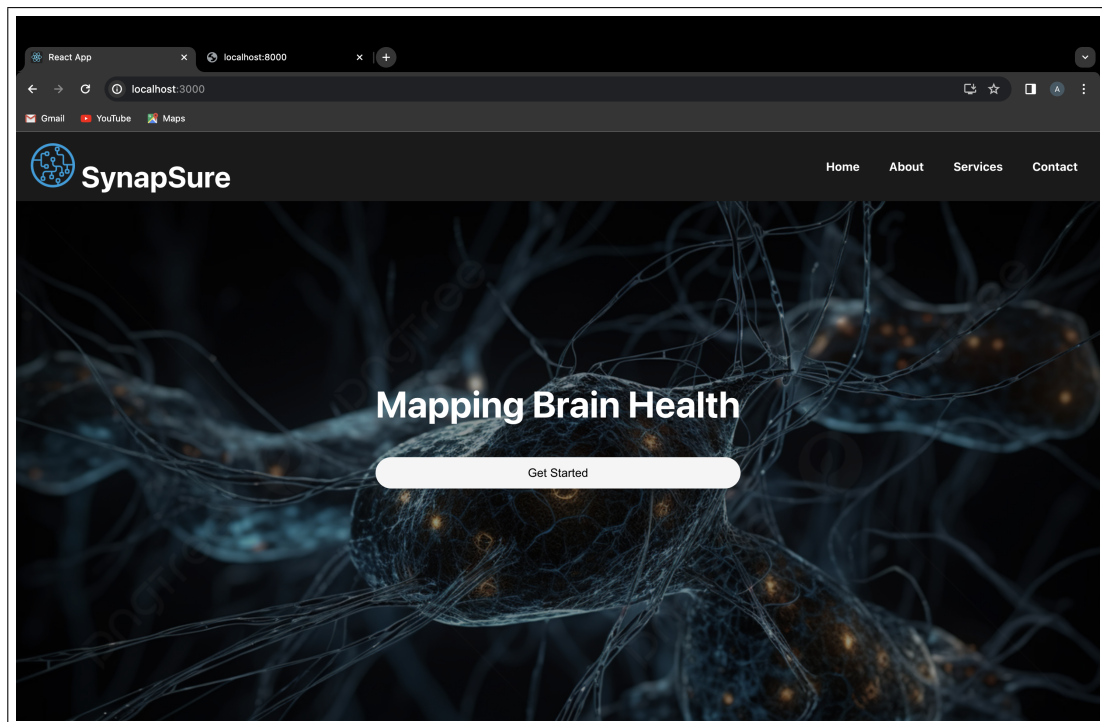


Figure 4.1: Landing Page

Landing page shows a navbar with options: Home, About, Services, and contact. Also, there is a button at the center of the page showing the option of "Get Started"

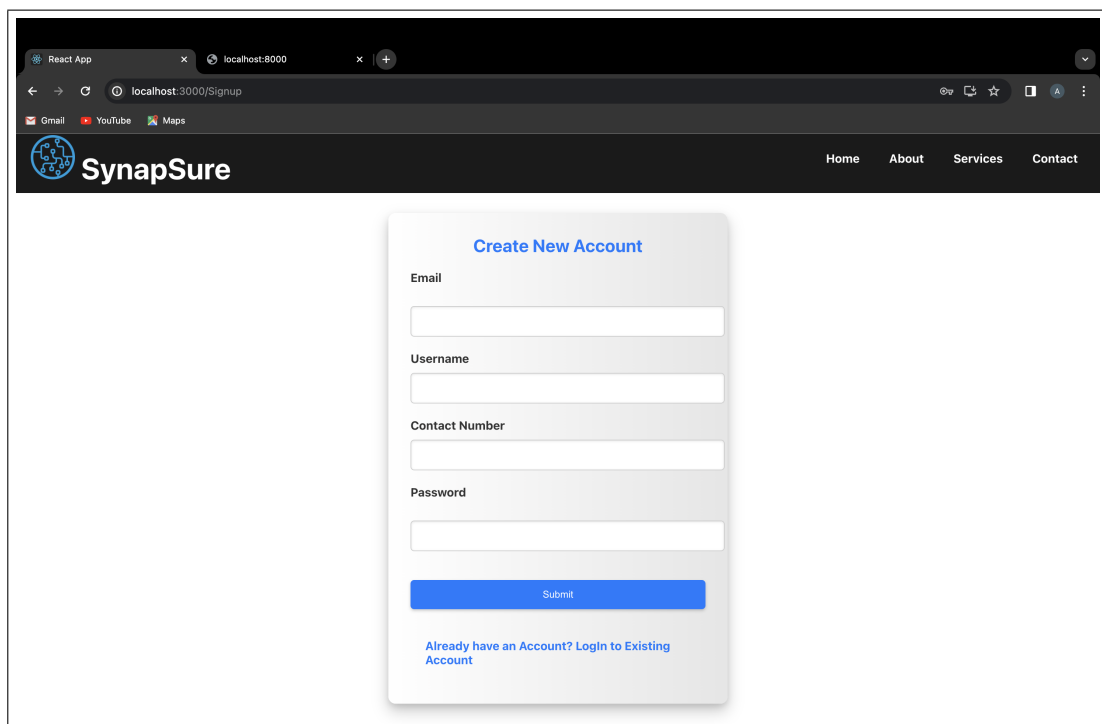


Figure 4.2: Sign-up Page

User friendly Sign-up Page for quick and easy registration. It includes the Email, Username, contact number and the password. There is also an option for already existing account

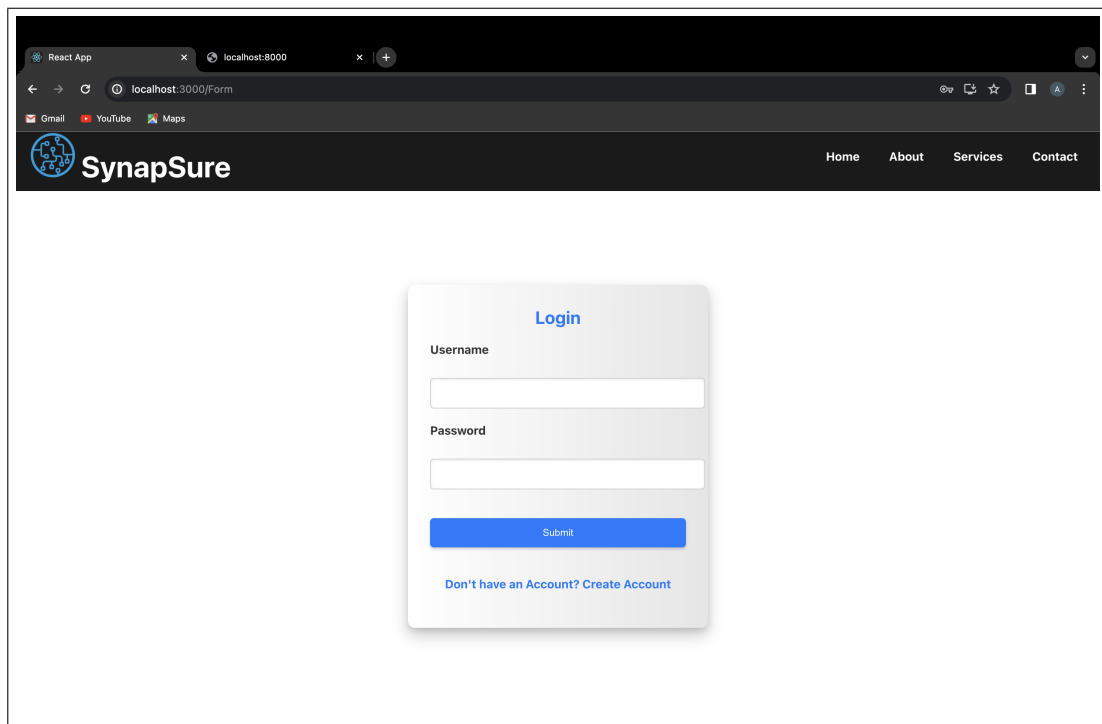


Figure 4.3: Login Page

Login Page for secure and easy access to SynapSure's dashboard. This figure includes the Username and password and there is also an option for creating a new account.

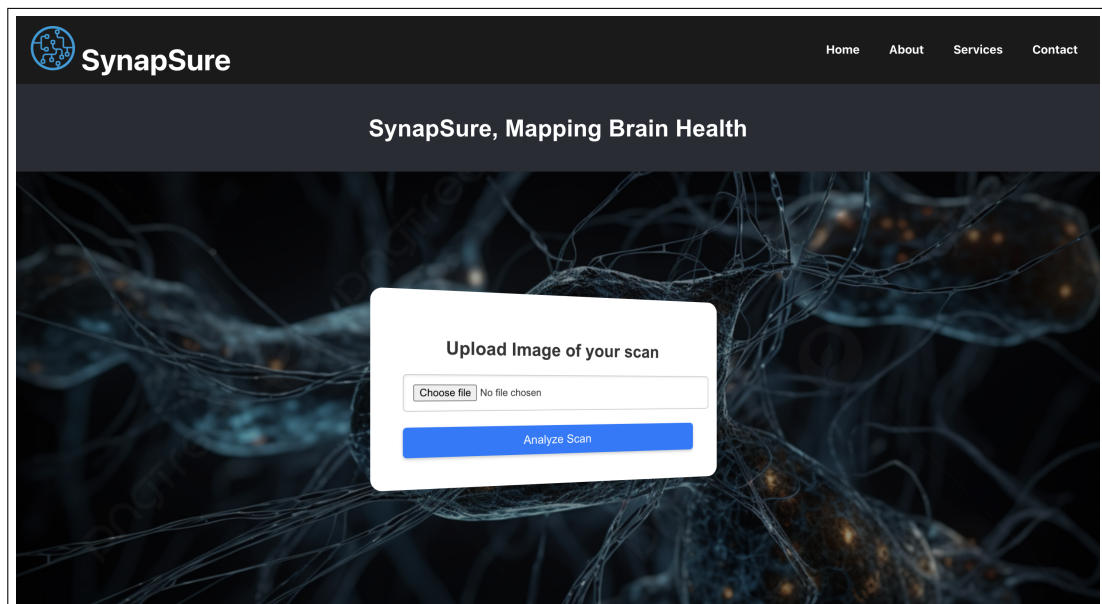


Figure 4.4: Upload Scan Page

Page for uploading brain scans allow user to upload their scan for analysis. This figure shows that the web page has a choose file button and then a button for analyzing the scan

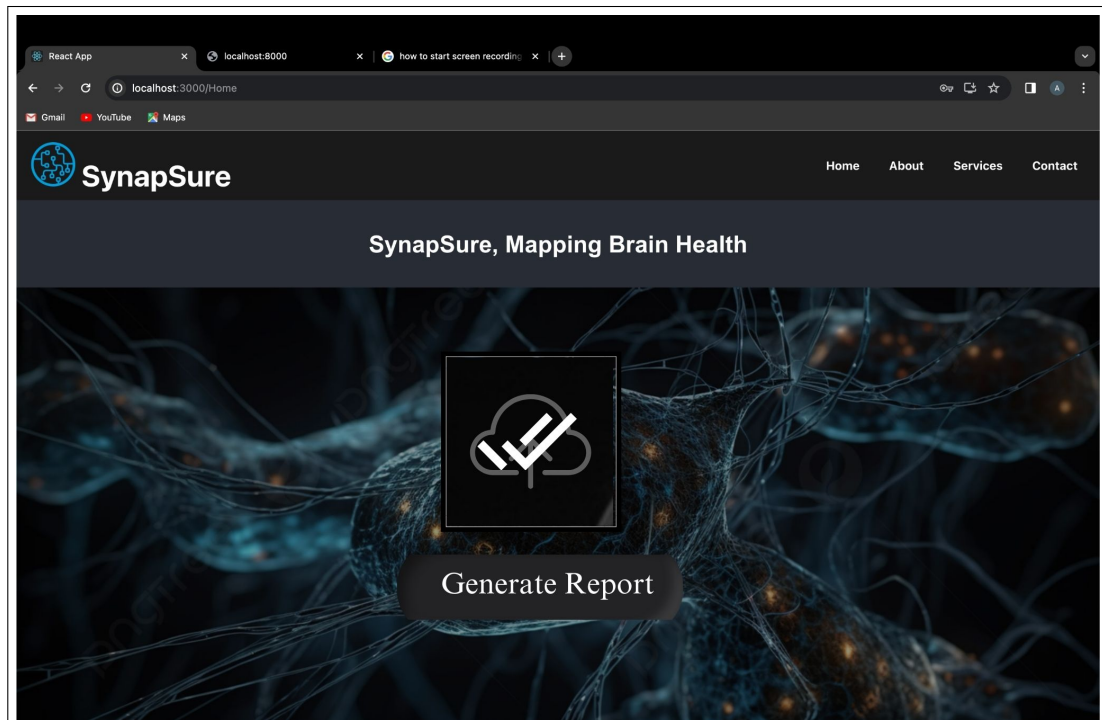


Figure 4.5: Generate Report Page

Page that allow user to generate analysis report by clicking on "Generate Report" button

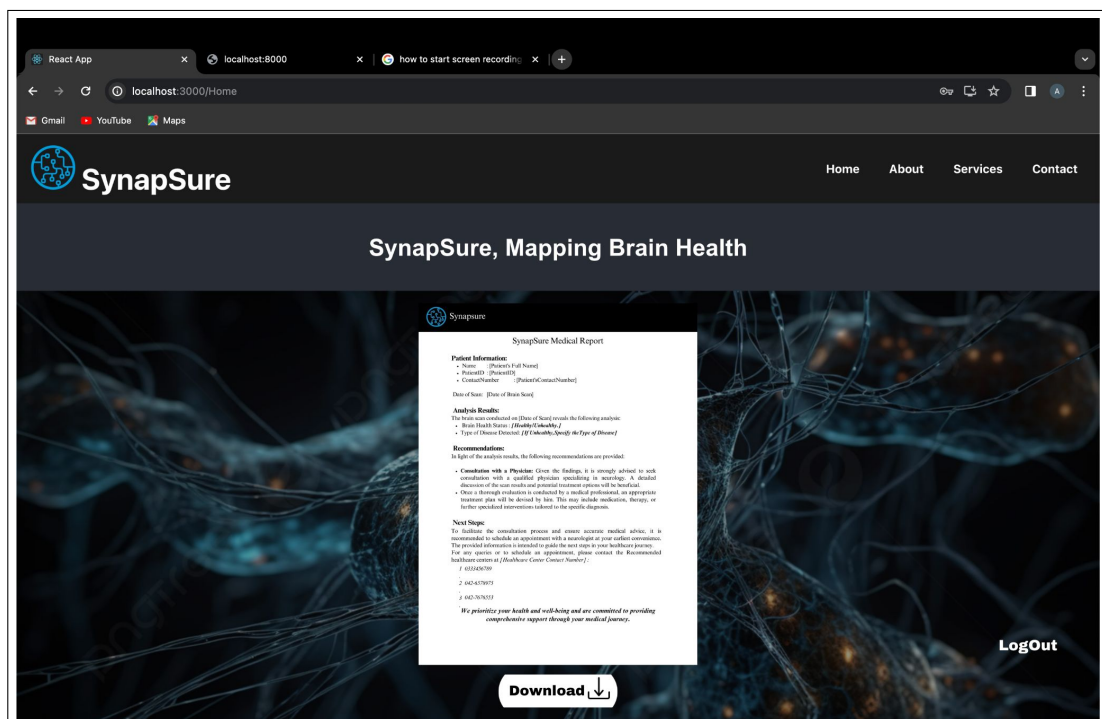


Figure 4.6: Report Page

Page displaying detailed diagnostic report and the option to download the report. It gives all the details of the patient and all details regarding the condition of brain



Synapsure

SynapSure Medical Report

Patient Information:

- Name : [Patient's Full Name]
- PatientID : [PatientID]
- ContactNumber : [Patient'sContactNumber]

Date of Scan: [Date of Brain Scan]

Analysis Results:

The brain scan conducted on [Date of Scan] reveals the following analysis:

- Brain Health Status : *[Healthy/Unhealthy.]*
- Type of Disease Detected: *[If Unhealthy,Specify theType of Disease]*

Recommendations:

In light of the analysis results, the following recommendations are provided:

- **Consultation with a Physician:** Given the findings, it is strongly advised to seek consultation with a qualified physician specializing in neurology. A detailed discussion of the scan results and potential treatment options will be beneficial.
- Once a thorough evaluation is conducted by a medical professional, an appropriate treatment plan will be devised by him. This may include medication, therapy, or further specialized interventions tailored to the specific diagnosis.

Next Steps:

To facilitate the consultation process and ensure accurate medical advice, it is recommended to schedule an appointment with a neurologist at your earliest convenience. The provided information is intended to guide the next steps in your healthcare journey. For any queries or to schedule an appointment, please contact the Recommended healthcare centers at *[Healthcare Center Contact Number]* :

- 1 0333456789
- .
- 2 042-6578975
- .
- 3 042-7676553
- .

We prioritize your health and well-being and are committed to providing comprehensive support through your medical journey.

Figure 4.7: Medical Report

Sample diagnostic report offering insights of the analysis results of brain scans performed by SynapSure. It includes the patient information which includes name and personal information. There is analysis results and recommendations and last is the next steps.

4.9 Database Design

4.9.1 ER Diagram

The relation of entities User, login, sign up and upload scan are shown with their relationship in crow foot notation such as a user can have one and only one account, a user can select zero or many scans to upload as shown in the figure below.

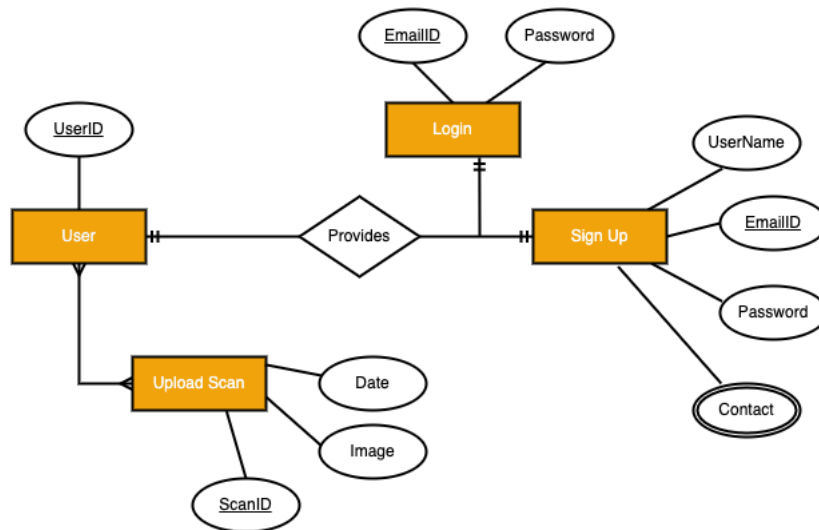


Figure 4.8: ER Diagram of SynapSure

Structure of the SynapSure project, the connections and dependencies between various data entities. It shows attributes for all

4.9.2 Data Dictionary

The data dictionary of the database, which will be used by the system, is shown below in the table.

Table 4.3: Data Dictionary

Overview of entities, attributes, data types, and descriptions in the database. There are five columns in this table

Entity	Attribute	Data Type	Nullable	Description
User	userID	Integer	No	The unique number assigned to the user.
Login	emailId password	String String	No No	The registered email address of the users. Password to authorize users.
Sign up	userName emailId password contact	String String String String	No No No Yes	User's full name. Unique email address of the user to register them in the system. Password to authorize users. The user's contact number which he can use for communication.
Upload Scan	scanId image date	int Image Date	No No No	The unique id of the image scan Image of the brain scan The date on which the brain was scanned.

4.10 Risk Analysis

As SynapSure aims for the diagnosis of brain scans it is crucial to conduct a comprehensive risk analysis to identify potential challenges and devise effective mitigation strategies. By anticipating and addressing these risks SynapSure can position itself for a successful and sustainable tool in brain diseases classification.

4.10.1 Technical Risks

The main technical risk that is considered concerns the availability and quality of brain scan data sets. For SynapSure to succeed there must be a wide array of large data sets for training and validation of machine learning algorithms. Biased or low sample data sets could lead to poor performing algorithms that will affect the precision with which brain conditions are classified. Further, with the advent of new machine learning and medical imaging technologies it is also possible that SynapSure's algorithms will become obsolete. It is necessary to always keep the latest methods of treatment as well as include them to SynapSure so that it remains significant and competitive.

4.10.2 Business Risks

SynapSure is an entity that operates in healthcare technology and in this domain, the rules, the market and competition dynamics change very fast. Changes on health policies or privacy rules might be introduced into the market of the project or other different diagnostic alternatives. Other emerging competitors such as other diagnostic techniques are significant business risks that require a good positioning and ongoing innovation in order to be ahead of them. Success of the project depends on effective cooperation with healthcare centers and any disturbances in this union could negatively influence the adoption and implementation of SynapSure.

4.10.3 Ethical and Privacy Risks

Since the medical information is very sensitive ethical and privacy issues dominate. It is important to the project that patients data be protected and they comply with privacy regulations. Breakdowns in data security will result to legal penalties degrade project image and lose trusts of users. To address ethical and privacy risks data protection techniques access controls and transparent data handling practices have to be implemented.

4.11 Conclusion

In this chapter, the software requirements for SynapSure are thoroughly discussed. The chapter provides a comprehensive road map for development by encompassing features, functional and non-functional requirements, quality attributes, and a graphical user interface. It emphasizes the need for reliability, maintainability, usability, and correctness which ensures the application's accuracy in medical contexts. Additionally, essential hardware and software requirements, user assumptions and comprehensive risk analyses addressing technical, business, and ethical concerns are also explained in the chapter. This information is like a guide for the next steps in making SynapSure a good and trustworthy tool for understanding brain diseases.

Chapter 5 Proposed Approach and Methodology

The SynapSure project adopts a modular approach consisting of Data Input, Preprocessing, Regression and Classification modules to achieve its goal of accurately diagnosing brain health conditions. This chapter provides detailed insights into each modules functionalities algorithms and significance within the system.

5.1 Data Input

The system initiates by acquiring diverse MRI brain scan data sets encompassing images and raw data from users. Leveraging the EfficientNet model the project reads and processes these scans. EfficientNet is selected for its efficient feature extraction capabilities crucial for accurate classification. These approaches ensures compatibility with multiple image format and seamless integration with subsequent module.

5.2 Data Preprocessing

Raw MRI data often contains noise that can hinder accurate classification. The Data Preprocessing module addresses this challenge using techniques like image normalization, filtering and enhancement. This module employs standardization of image sizes noise reduction through filters and enhancement of relevant features. Traditional image processing and deep learning methodologies are combined to optimize data quality.

5.3 Regression

To differentiate between healthy and unhealthy brain scans the Regression module is introduced. Utilizing regression analysis the system assesses quantitative indicators within the scans and establishing a baseline for health. Regression helps identify patterns and deviations associated with unhealthy conditions which establishes the groundwork for subsequent classification. The regression model is trained on a diverse data set to capture a broad spectrum of health variations.

5.4 Classification

Following regression analysis the project utilizes the EfficientNet model for further classification of unhealthy brain scans. The model pre trained on extensive data sets ensures adaptability and accuracy in identifying specific conditions like tumors, strokes and Alzheimer's. Transfer learning is employed to

overcome data scarcity and hence enhancing the model's precision. The results are presented through a user-friendly interface, providing users with comprehensive insights into their brain health.

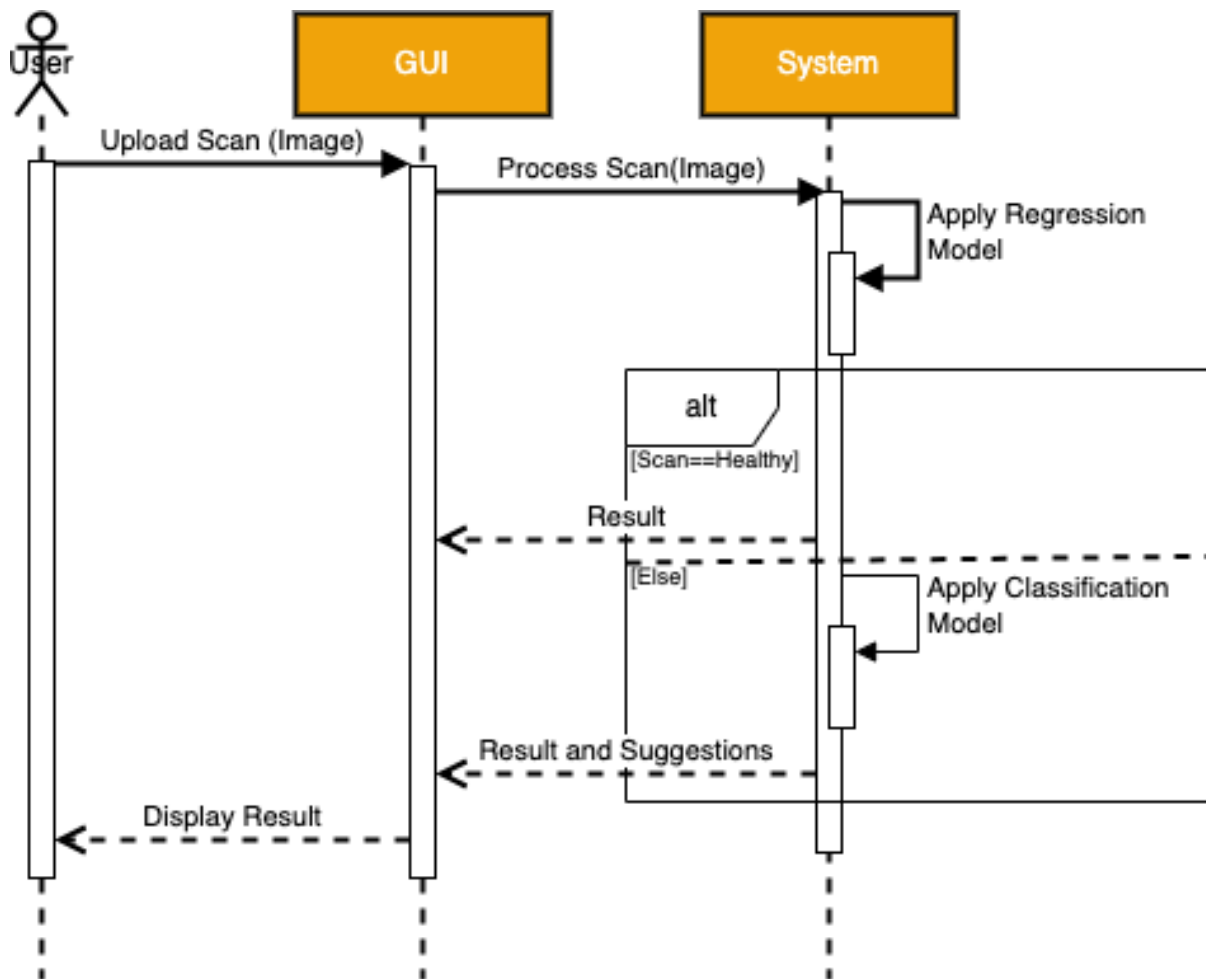


Figure 5.1: Sequence Diagram of Complete System

Visual representation depicting the sequence of interactions within the complete system. This figure shows the steps which includes input scan and then process the scan and then the regression model and after classification and then it displays the results.

5.5 Conclusion

In conclusion the proposed approach and methodology for SynapSure emphasizes a systematic process from data input to regression and classification. By integrating advanced models like EfficientNet for classification and introducing regression for initial differentiation the system ensures adaptability and accuracy in diagnosing a spectrum of brain health conditions. The modular design allows for scalability and future enhancements aligns with the project's commitment to providing a reliable, user friendly and precise brain health diagnostic tool.

Chapter 6 High-Level and Low-Level Design

6.1 System Overview

In this chapter we explore the comprehensive design of SynapSure a revolutionary medical diagnostic platform that integrates machine learning techniques for improved brain health assessments.

6.1.1 System Design

The SynapSure system architecture incorporates a number of crucial steps that provide an easy move from collecting data to real time analysis.

6.1.1.1 Data Acquisition and Preprocessing

The procedure begins with the collection of various brain scan data. SynapSure accurately classify disorders such as brain tumors, strokes, and hemorrhages by using pre trained models. Data preprocessing is the process of extracting useful information from unprocessed data and improving input.

6.1.1.2 Model Development

The careful development of advanced machine learning models is the foundation of SynapSure. The EfficientNet model a customized convolutional neural network serves as a foundational part which is capable of extracting features required for accurate brain health analysis. SynapSure includes regression models with CNNs providing an additional layer to the systems analytical abilities. This integration allows for more complete evaluation and prediction strengthening the platform's ability to spot abnormalities and measure overall brain health. Through the synergy of CNNs and regression models, SynapSure achieves a harmonious blend of pattern recognition and quantitative analysis. This dual-model approach ensures a robust framework for early detection and precise treatment planning underscoring SynapSure's commitment to advancing neurology diagnostics for improved patient outcomes.

6.1.1.3 Real Time Assessment

SynapSure makes sure users can use the platform smoothly. The interface is designed to help detect problems early which is important for making sure patients get better results in their treatment. This explains that users can quickly and accurately understand what is happening with the brain scans which makes the whole process more effective and helpful.

6.1.2 Data Collection

The SynapSure dataset is gathered with great care in controlled settings which makes sure that it has the best examples for training and testing the machine learning models. This information shows how users behave within the specific limits of the project. The datasets include almost 42,000 brain scans for brain tumor, 12,000 brain scans for brain stroke and around 60,000 brain scans for brain hemorrhage. Having such a diverse and large dataset helps the models learn well and become really good at identifying different brain conditions accurately.

6.1.3 Model Training

During SynapSure model training phase we employ the complicated EfficientNet model with the regression algorithms. For effective utilization of deep learning capabilities in precise and specific brain health diagnosis this strategic choice cannot be overemphasized. The EfficientNet model which is well known for its superior performance in image classification tasks is good at identifying minute trends in a wide range of brain imaging data. The regression techniques help enhance EfficientNet model by providing with continuous readings that help give deeper insight into brain health measures. SynapSure uses this dual approach that combines the strengths of EfficientNets model and regression in order to provide precise and informative assessment of serious conditions such as brain tumor, stroke and hemorrhage. SynapSure aims to optimize the EfficientNet model and regression algorithms by training on a large data set ensuring the system's reliability, adaptability and effectiveness in changing the brain health diagnosis.

6.2 Design Considerations

This section addresses critical issues and considerations essential for SynapSure's development.

6.2.1 Assumptions and Dependencies

6.2.1.1 Assumptions

- The system consider that the collected data is reliable and free from errors.
- SynapSure relies on the proper operation of hardware components such as hardware devices in order to run our website.

6.2.1.2 Dependencies

- SynapSure relies on users being available for brain imaging evaluations to upload the scan.

- SynapSure dependent on internet to upload scans on our website and to generate its report.

6.2.2 General Constraints

SynapSure recognizes limitations that impact its software design such as hardware limits and data standards compliance and performance needs.

6.2.2.1 Key Constraints

- The specs and capabilities of the hardware used for brain imaging restrict system performance.
- Compatibility with defined data standards is critical for scalability and correct operation of system.

6.2.3 Goals and Guidelines

6.2.3.1 Key Objectives

- SynapSure aims for high precision in brain health assessments ensuring accurate classification of conditions.
- The system is designed to be adaptable accommodate future changes in machine learning for effectiveness.
- SynapSure depends on scalability to handle large volumes of brain imaging data and improves diagnostic requirements.

6.2.4 Development Methods

SynapSure uses the Scrum development process which is a flexible and step-by-step way to build and improve the software. With Scrum we can keep adding new features smoothly and adjust things easily if needed. It is like a continuous process of making the software better bit by bit. This makes SynapSure really adaptable and able to meet changing needs. It is like making sure the software is always ready and works well with the latest improvements which creates a system that is both active and good at what it does.

6.3 System Architecture

SynapSure's architecture provides a high level overview of its functionality.

6.3.1 Subsystem Architecture

6.3.1.1 Major Components

- Collects the diverse brain imaging data from the various sources.
- Involves creation and training of machine learning models for the accurate classification.
- Enables users to interact with the platform hence contributing to real time assessments.

6.4 Architectural Strategies

SynapSure employs various strategies to shape its architecture and the decision making processes.

6.4.1 Technology Stack

6.4.1.1 Programming Language

In the backend development of SynapSure we use the Python language which is great because it works well with machine learning libraries and frameworks. For the frontend development we use React a popular technology that helps create a user friendly and interactive interface. This combination of Python for the backend and React for the frontend ensures that SynapSure is equipped with the right technologies to provide a smooth and effective user experience.

6.4.1.2 Database

MongoDB is chosen for its flexibility and scalability in storing and retrieving brain imaging data.

6.4.1.3 Machine Learning Framework

In our methodology we have adopted the EfficientNet model for its superior performance coupled with a regression approach as outlined in the previous chapter. TensorFlow which a powerful and versatile machine learning framework is the engine behind our model development and training. Its robust capabilities play a pivotal role in ensuring the efficiency and effectiveness of our machine learning endeavors.

6.4.2 Scalability and Adaptability

SynapSure's architecture is designed to scale seamlessly accommodating an increasing volume of brain imaging data. The system is built to be adaptable, allowing for the integration of future advancements in machine learning technologies.

6.5 Domain Model/Class Diagram

Following is the model diagram of the project SynapSure.

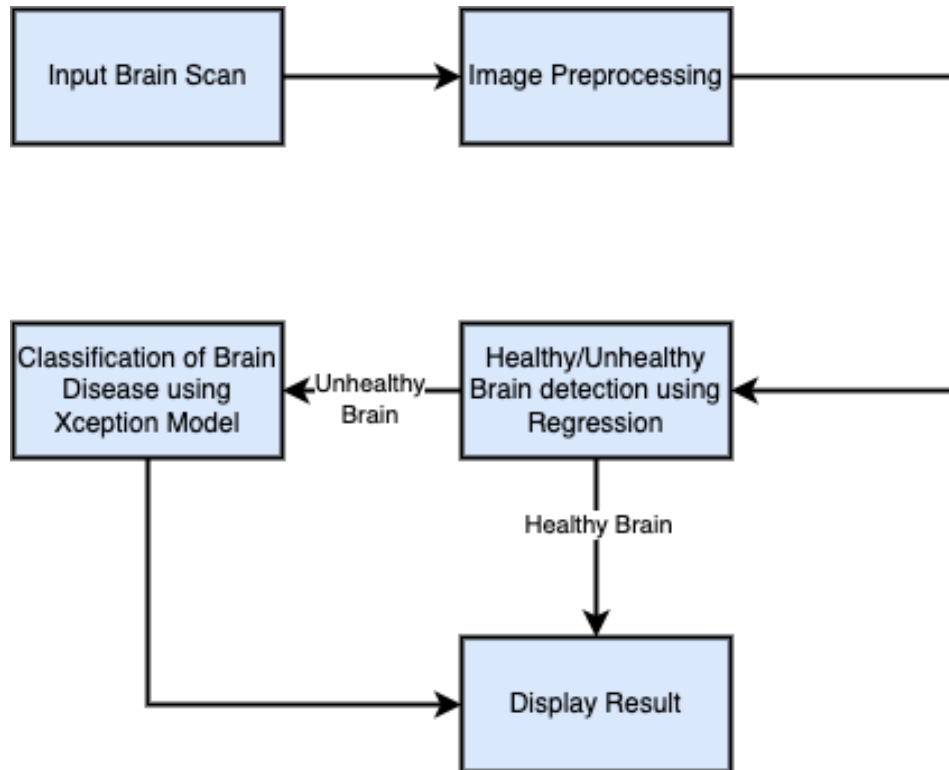


Figure 6.1: Model Diagram of SynapSure

A visual representation illustrating the overall model architecture of SynapSure. This figure shows the steps of model. It starts from inputting the brain scan and then image processing and then regression is done and then it classifies brain disease using EfficientNet model and then it displays the result

The above diagram explains the proposed system. The model begins with the input of brain scans which undergo a image preprocessing stage aimed at enhancing data quality through techniques like normalization and noise reduction etc. The subsequent stages involve feature extraction and selection where relevant patterns and structures in the brain images are identified and isolated for further analysis. The major point in the model is the classification stage, which aims to determine whether the scanned brain is healthy or unhealthy. If the brain is deemed healthy the system displays the result. In cases where unhealthy brain is identified the system further delves into disease classification, employing advanced algorithms to detect specific conditions such as tumors, strokes and hemorrhage. The final step involves displaying the detailed results and suggestions. In summary this model it ensures a comprehensive and effective assessment of brain scans which provides medical professionals with valuable insights into the complex domain of brain health.

6.6 Sequence Diagrams

Following are sequence diagrams that are presented in our web application.

6.6.1 SignUp

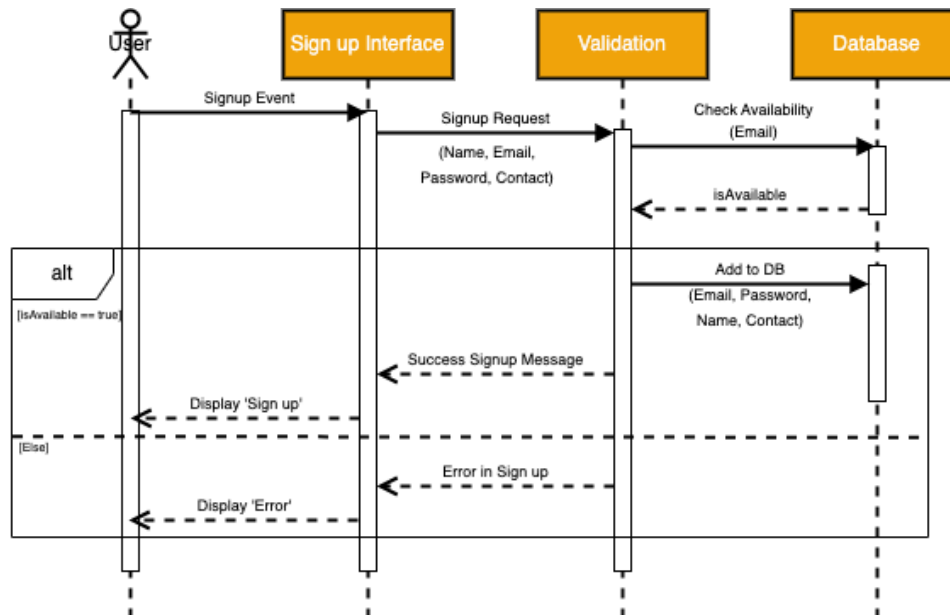


Figure 6.2: SignUp

This figure represents the Signup Sequence Diagram of our system. It shows all the steps starting from sign up event and then request of sign up and it displays the sign up or error

6.6.2 Login

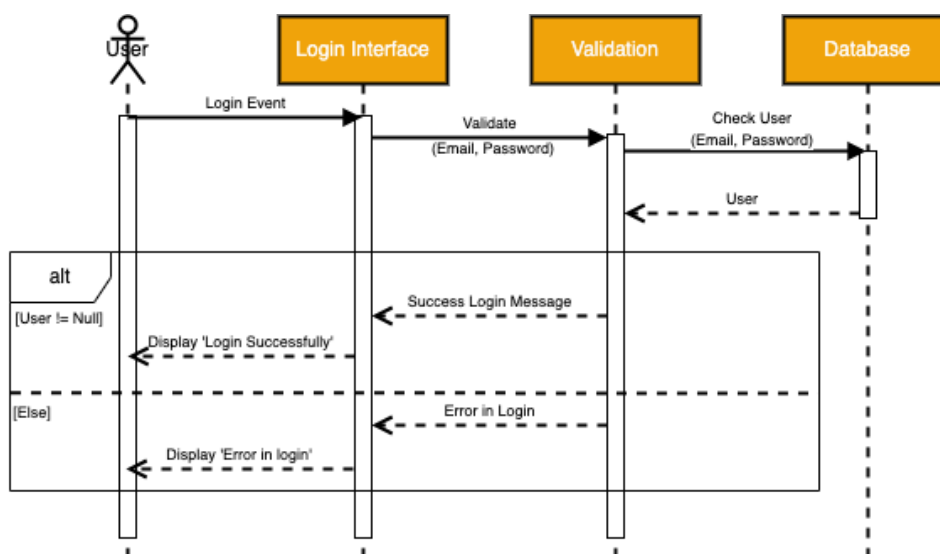


Figure 6.3: Login

This figure represents the Login Sequence Diagram of our system. It starts from the login event and then it checks the user and ends with displaying user login successfully or it displays error

6.6.3 Upload Scan

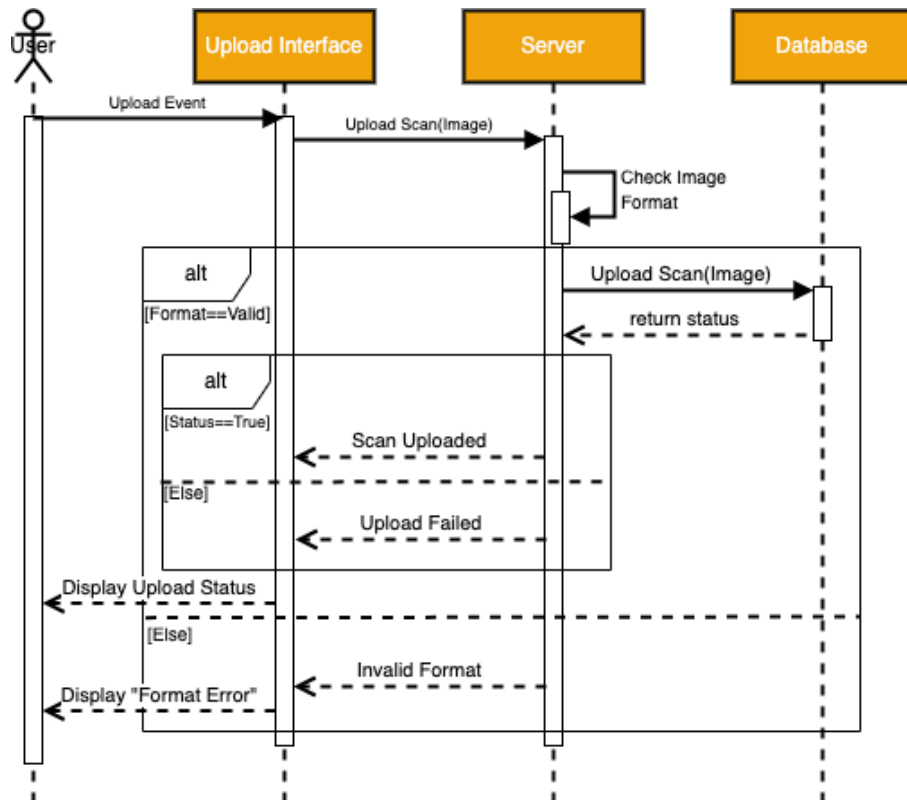


Figure 6.4: Upload Scan

This figure represents the Upload Scan Sequence Diagram of our system. It starts from uploading the event and then upload scans and after processing it shows scan uploaded or upload failed

6.7 Policies and Tactics

This section outlines specific policies and tactics influencing interface and the implementation details of SynapSure.

6.7.1 Software Tools

- SynapSure employs Visual Studio Code and Google Colaboratory for efficient development, testing and deployment.
- Git is used for tracking the changes and collaborating on the code base.

6.7.2 Coding Standards

Coding standards in Python are followed to maintain code readability and consistency. Regular code reviews are conducted to ensure the adherence to the coding guidelines.

6.7.3 Testing Procedures

SynapSure follows a rigorous testing process including unit testing, integration testing and user acceptance testing. Continuous testing ensures the reliability and accuracy of the machine learning models.

6.7.4 Software Maintenance

Regular updates and maintenance procedures are implemented, to keep SynapSure's software up-to-date. Future enhancements including the incorporation of new machine learning algorithms are planned to improve diagnostic capabilities.

6.8 Conclusion

This chapter has provided a detailed exploration of the high level and low level design considerations for SynapSure. From system architecture to design goals and development methodologies the chapter highlights the thought process and decisions, guiding the development of this innovative medical diagnostic platform. The subsequent chapters will delve into the detailed implementation and testing phases hence providing a comprehensive understanding of the SynapSure project.

Chapter 7 Implementation and Test Cases

The implementation of SynapSure involves advanced machine learning models especially EfficientNet model for image classification using transfer learning and regression for binary classification. This chapter provides a detailed overview of the implementation process algorithms used and frameworks and tools employed.

7.1 Implementation

7.1.1 EfficientNet Model for Image Classification

A pre-trained convolutional neural network model, EfficientNet is employed for the classification of brain scans into three categories which are tumor, stroke, and hemorrhage. Transfer learning is implemented to maximize the EfficientNet model's learning obtained from a large data set. The following steps outline the implementation:

7.1.1.1 Transfer Learning

Transfer learning involves using a model that has already learned from one dataset and applying it to a new one. In the case of SynapSure, we employ the EfficientNet model which initially trained on a general dataset. After that we fine-tune this model specifically for our brain scan dataset. This means the model can utilize the knowledge gained from the larger dataset to better recognize different brain conditions in our specific scans which enhances accuracy and effectiveness in identifying abnormal brain conditions.

7.1.1.2 Fine-Tuning

Since our brain scan data set is relatively small, fine-tuning is necessary to adjust the pre-trained EfficientNet model to our specific features. We freeze the initial layers of the EfficientNet model and allow only the later levels to be fine-tuned during training. This approach prevents over fitting and ensures that the model captures relevant characteristics from our dataset which increases its ability to identify different brain conditions.

7.1.2 Regression for Binary Classification

For binary classification, regression is employed to differentiate between normal and abnormal brains. Implementation details are as follow:

7.1.2.1 Data Preprocessing

To increase the model's ability to categorize brain scans we performed preprocessing on the images. This involves tasks such as resizing and normalization which make sure that the images are in a consistent and optimal format for analysis. These processes help train the model to be more robust and adaptable which improves its accuracy in identifying different brain conditions. It is related to preparing the images for the model to learn and make better predictions based on variations. In addition to the preprocessing steps we aimed at enhancing the categorization capabilities of our brain diseases classification machine learning model, we are using partially annotated dataset. This dataset contains brain scan images that have undergone some level of annotation, though not comprehensively labeled across all relevant categories. Despite the partial annotations, the dataset offers valuable insights into various brain conditions which helps the training of our model. This approach enables the model to learn from the available annotations while also accommodating potential variations in the data, which improves its accuracy in classifying different brain conditions.

7.1.2.2 Regression Model

TensorFlow and Keras are used to create a regression model capable of understanding patterns in pre-processed brain scan images. The goal is to classify whether the scanned image is normal or abnormal. Through training on a dataset of preprocessed brain scans, the model learns to distinguish between different conditions which contributes to the accurate identification of abnormalities in future brain scans. This approach certify that the model is well-equipped to make reliable assessments based on the learned information.

7.1.2.3 Integration with EfficientNet Model

The regression model results support the EfficientNet model classifications which provides an extensive overview of brain disorders.

7.2 Test case Design and description

This section include input constraints that are true for every input in the set of associated test cases, any shared environmental needs, any shared special procedural requirements, and any shared case dependencies.

Table 7.1: Test case for Sign Up

Signup			
1			
Test Case ID:	1	QA Test Engineer:	Fatima Muneer
Test case Version:	1.0	Reviewed By:	Awais
Test Date:	30-04-2024	Use Case Reference(s):	4.6.1 Sign Up
Revision History:	None		
Objective:	To check if user is able to successfully signup to website		
Product/Ver/Module:	Website signup module		
Environment:	Website is running on web browser and internet is connected		
Assumptions:	Valid email address and secure password		
Pre-Requisite:	User must not exists in the database		
Step No.	Execution description	Procedure result	
1.	User enters the sign up details i.e. name, email, password, phone number	Passed	
2.	System stores the user's information to database	Passed	
3.	System takes the user to login page	Passed	
Comments: Test case passed. The system works according to our requirements			
Passed			

Table 7.2: Test case for Login

Login			
1			
Test Case ID:	2	QA Test Engineer:	Fatima Muneer
Test case Version:	1.0	Reviewed By:	Akbar Khan
Test Date:	30-04-2024	Use Case Reference(s):	4.6.2 Login
Revision History:	None		
Objective:	To check if user is able to successfully login to the website		
Product/Ver/Module:	Website login module		
Environment:	Website is running on web browser and internet is connected		
Assumptions:	Valid email and password		
Pre-Requisite:	User must exists in the database		
Step No.	Execution description	Procedure result	
1.	User selects the login option	Passed	
2.	User inputs valid email and password	Passed	
3.	System takes the user to home page	Passed	
Comments: Test case passed. The system works according to our requirements			
Passed			

Table 7.3: Test case for Upload Brain Scan

Upload Brain Scan			
2			
Test Case ID:	3	QA Test Engineer:	Awais
Test case Version:	1.0	Reviewed By:	Fatima Muneer
Test Date:	30-04-2024	Use Case Reference(s):	4.6.3 Upload Brain Scan
Revision History:	None		
Objective:	To check if user is able to successfully upload their brain scan to the website		
Product/Ver/Module:	Upload Scan module		
Environment:	Website is running on web browser and internet is connected		
Assumptions:	High resolution scan images		
Pre-Requisite:	User is logged in to the website and scan is in supported format		
Step No.	Execution description	Procedure result	
1.	User clicks the upload scan option	Passed	
2.	User choose the image from device	Passed	
3.	System stores the scan to database	Passed	
Comments: Test case passed. The system works according to our requirements			
Passed			

Table 7.4: Test case for View Analysis Result

View Analysis Result			
3			
Test Case ID:	4	QA Test Engineer:	Awais
Test case Version:	1.0	Reviewed By:	Akbar Khan
Test Date:	30-04-2024	Use Case Reference(s):	4.6.4 View Analysis Result
Revision History:	None		
Objective:	To check if user can access the analysis report		
Product/Ver/Module:	Analysis Result module		
Environment:	Website is running on web browser and internet is connected		
Assumptions:	-		
Pre-Requisite:	Brain scan has been analyzed by the system		
Step No.	Execution description	Procedure result	
1.	User clicks the "view scan" option for the specified scan	Passed	
2.	System displays the analysis result	Passed	
Comments: Test case passed. The system works according to our requirements			
Passed			

Table 7.5: Test case for Provide Feedback

Provide Feedback			
4			
Test Case ID:	5	QA Test Engineer:	Akbar Khan
Test case Version:	1.0	Reviewed By:	Fatima Muneer
Test Date:	30-04-2024	Use Case Reference(s):	4.6.5 Provide Feedback
Revision History:	None		
Objective:	To check that user can provide feedback on their experience		
Product/Ver/Module:	Feedback module		
Environment:	Website is running on web browser and internet is connected		
Assumptions:	-		
Pre-Requisite:	User is logged in to the website		
Step No.	Execution description	Procedure result	
1.	User selects the "Provide feedback" option	Passed	
2.	User fills the form and submit it	Passed	
3.	System stores the feedback to database	Passed	
Comments: Test case passed. The system works according to our requirements			
Passed			

Table 7.6: Test case for Provide Feedback

Generate Report			
5			
Test Case ID:	6	QA Test Engineer:	Akbar Khan
Test case Version:	1.0	Reviewed By:	Awais
Test Date:	30-04-2024	Use Case Reference(s):	4.6.6 Provide Feedback
Revision History:	None		
Objective:	To generate a detailed diagnosis report		
Product/Ver/Module:	Generate Report module		
Environment:	Website is running on web browser and internet is connected		
Assumptions:	-		
Pre-Requisite:	User is logged in to the website and brain scan has been analyzed		
Step No.	Execution description	Procedure result	
1.	User selects the "Generate Report" option	Passed	
2.	System displays the report	Passed	
3.	User can view or download the report	Passed	
Comments: Test case passed. The system works according to our requirements			
Passed			

7.3 Test Metrics

Summarize here the common ground of attributes of test case metrics as explained in Table 7.7.

Table 7.7: Test cases Matric

Following are the test metrics and their results after the testing process

Metric	Purpose
Number of Test Cases	6
Number of Test Cases Passed	6
Number of Test Cases Failed	0
Test Case Defect Density	0
Test Case Effectiveness	100
Traceability Matrix	-

7.4 Conclusion

In conclusion, SynapSure is implemented using transfer learning with the EfficientNet model for image classification and a regression model for binary classification. This section provides a detailed overview of the implementation of the desired algorithms and emphasizes the necessary frameworks. EfficientNet was chosen for image classification due to its ability to benefit from transfer learning and efficiently categorize brain scans into various unhealthy categories, especially when utilizing a partially annotated dataset. Data preprocessing remains a crucial step to ensure the accuracy of the model's predictions, facilitating tasks such as resizing and normalization. Integrating a regression model with the EfficientNet model further enhances the classification and analysis of brain disorders, offering a comprehensive approach to understanding and categorizing brain conditions.

Chapter 8 User Manual

8.1 Installation and Setup

To begin using SynapSure, users must download and install the application (.exe file).

8.1.1 Features

8.1.1.1 Buttons

SynapSure provides the following buttons for user interaction:

Get Started: Initiates the SynapSure application.

Log In: Allows the user to log in to the application.

Upload Image: Enables the user to upload an image for analysis.

Generate Report: Triggers the generation of the analysis report.

Download Report: Allows the user to download the generated report.

Log Out: Enables the user to log out of the SynapSure application.

8.1.2 Application Workflow

The workflow for using SynapSure is as follows:

User clicks on "Get Started": This initiates the SynapSure application.

User logs in: The user provides their credentials to access the application.

User uploads the image: Upon logging in, the user uploads the image for analysis using the "Upload Image" button.

User generates the report: After uploading the image, the user generates the report containing the analysis results by clicking the "Generate Report" button.

User downloads the generated report: Once the report is generated, the user can download it for further reference using the "Download Report" button.

User can log out: Finally, the user can choose to log out of the SynapSure application using the "Log Out" button.

Chapter 9 Experimental Results and Discussion

In this chapter, we conducted experiments using small data. We present the results for shortlisted models after experimenting.

Table 9.1: Model Comparison Results

The table shows the results of the all three tested models, Efficient model, Xception model and resnet model, which we compared to find most suitable model for our project to classify brain scans.

Dataset	Accuracy (%)
Efficientnet Model	99%
Xception Model	82%
Resnet Model	81%

9.1 EfficientNet Model Results

The EfficientNet model was able to classify abnormal brain scans into particular diseases with encouraging results. The model had approximately 99% accuracy on a diverse dataset. We can fine-tune our model through further refining with larger training sets that ultimately leads to increased precision in later iterations of this process.

9.2 Xception Model Results

The implementation of the Xception model demonstrated substantial success in classifying abnormal brain scans into specific diseases and achieves an accuracy of 82% on a diverse dataset. Further optimization through fine-tuning with larger training sets is planned to refine the model's performance and aims for increased precision in subsequent iterations of the classification process.

9.3 ResNet Model Results

In the utilization of the ResNet model the commendable outcomes were achieved in the classification of abnormal brain scans and attains an accuracy of 81% on a diverse dataset. The ongoing objective involves fine-tuning the model with larger training sets to enhance precision in subsequent iterations that showcasing the potential for improved accuracy in identifying different brain conditions.

9.4 Handling Outliers and Unknown Class

We utilized outlier detection algorithms and strategies for handling scans which belong to an "Unknown" class to increase the robustness of the SynapSure system. To identify and handle outliers in the dataset an

outlier detection approaches such as statistical measures clustering and isolation forests were utilized. We used a threshold-based classification strategy for datasets that did not fall into any of the specified classifications. The regression and EfficientNet models aimed to generate a confidence score and datasets with confidence less than a certain threshold were classified as "Unknown". This method ensures that the system can handle outliers and cases that do not fit into the specified classification groups. Continuous learning and frequent model updates let the system adapt to new data patterns.

Table 9.2: Outlier Handling and Unknown Class Classification

This table delineates the sequential phases and methodologies encompassed in the process of handling outliers and classifying scans into an "Unknown" category. The table consists of three distinct columns: "Phase" signifies the step number in the process, "Outlier Handling Steps" elucidates the actions undertaken to manage outliers, and "Methods" outlines the specific techniques employed at each phase.

Phase	Outlier Handling steps	Method
1.	Identify Outliers	Statistical measures, clustering, or isolation forests are used to detect outliers in the dataset.
2.	Set Threshold	Based on the selected outlier detection method, a threshold is selected to indicate the amount of variation beyond which a brain scan can be considered an outlier.
3.	Handle Outliers	Outliers can be removed from the training dataset or transformed to reduce their impact on model training.
4.	Continuous Learning	The system incorporates continuous learning, updating models and thresholds based on new data to improve outlier detection accuracy.
5.	Confidence Score	Models provide a confidence score for each prediction, indicating the model's certainty about the classification.
6.	Dynamic Threshold	A dynamic threshold is set for the confidence score, adapting to changing conditions and uncertainties.
7.	Categorize as "Unknown"	Brain scans with confidence scores below the threshold are categorized as belonging to the "Unknown" class.
8.	Adaptive System	The system continuously adapts to new patterns, ensuring flexibility and improved accuracy in handling the "Unknown" class.

9.5 Discussion

The achieved accuracy highlights the SynapSure project's efficiency in classifying brain images. In comparing the accuracies of three prominent models, EfficientNet demonstrated superior performance with an accuracy of 99%, overtaking both Xception (82%) and ResNet (81%). The selection of EfficientNet for the SynapSure project was motivated by its robust performance which provides a balance between

accuracy and computational efficiency. The model's ability to handle diverse datasets and its fine-tuning potential with larger training sets make it a suitable choice for optimizing precision in the identification of various brain conditions. The integration of EfficientNet along with reliable outlier identification and an "Unknown" class handling method contributes to the reliability and effectiveness of SynapSure in the classification of brain scans which aligns with the goal of project of providing a powerful tool for healthcare professionals in diagnosing and understanding brain conditions.

9.6 Future Enhancements

The future goals of SynapSure will be to refining the outlier detection algorithms and improving methods for handling an "unknown" class, SynapSure's future goals include leveraging DC-GAN to generate more synthetic data. This approach will enhance the diversity and quantity of the dataset which helps in better classification performance. Furthermore, SynapSure will implement a regression model to classify whether the input brain scan is of a healthy or unhealthy brain. Next versions of the system will make sure to keep the data diversified and updated so that it can be accurate and adaptable to real-world applications. The main goal of SynapSure is to consistently improve its capabilities in classifying brain scans and diagnosing them so users can get accurate results.

9.7 Conclusion

This chapter concludes the important results and discusses the main topic of the SynapSure project. Regression models for classification and EfficientNet model to categorize the abnormalities of brain scans were implemented by testing different datasets (with small datasets). In the results from the EfficientNet model, promising classification among diseases like brain tumors, strokes, and hemorrhages was achieved. The system shows its strength and learning capabilities from new data by managing unknown classes and different techniques to handle outliers within the dataset. Through this study, we can draw conclusions for and assure the reliability to use SynapSure as a classifier in imaging brain scans efficiently.

Chapter 10 Conclusions

Our project uses modern machine learning algorithms to classify brain conditions, in an interesting and comprehensive way. SynapSure uses regression for binary classification and integrates transfer learning, with the EfficientNet model to accurately classify brain scans into several categories such as normal, tumor, stroke and hemorrhage. This research highlights the value of fine-tuning and pre-trained models by fitting them to distinct data sets in addition to emphasizing the use of modern technology in medical imaging. By offering a reliable and efficient tool, for identifying and comprehending brain disorders SynapSure intends to further the fields of neuro imaging and the medicine.

In our experimental results we expect high accuracy, particularly when using the pre-trained EfficientNet model on brain images. However the challenge of handling abnormal brain scans, that do not fall into the predefined classes requires notable consideration. To address this we are exploring outlier detection and unknown class handling methods hence ensuring the model's adaptability to a broader spectrum of brain conditions. In addition to the research, a user-friendly graphical user interface has been created. All design considerations, goals, and guidelines that are a part of this web application have been while keeping the basic functionality in mind.

The project aims to conduct extensive experiments to validate the effectiveness of the implemented algorithms. This includes increasing the model's accuracy such as fine tuning the EfficientNet model for optimal performance. We aim to provide a detailed analysis of the experimental results, discussing both successes and areas for improvement. We pointed out the important role of annotated data in enhancing the performance of our models. Annotated data allowing algorithm to learn and generalize the patterns correctly. Through using annotated data, we are promoting robustness and accuracy of our algorithms in the detection of brain conditions of various types. Additionally, we will incorporate larger datasets to increase model accuracy using different techniques like outlier detection. Moreover, we aspire to integrate the trained machine learning model with the front-end using different APIs creating a fully functional website. MongoDB will serve as the database to store all information on our website, ensuring a seamless and comprehensive user experience. Through the collaborative efforts of machine learning and medical imaging, SynapSure strives to contribute to the field of neuro imaging, offering a reliable tool for healthcare professionals in diagnosing and understanding various brain conditions.

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